

U-Jagd Manual

Player manual, version 1.3.114, as of 2026-09-30

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Quickstart

You command the ASW frigate F-217 and man nine stations. Your job: detect, track, classify and sink hostile submarines without losing the ship or harming neutral shipping.

Everything in this manual describes what the simulation actually models. Features that are not modelled are listed at the end of every station chapter.

Mission and win conditions

- **Win:** sink every assigned target submarine, or survive until the time limit (depends on the mission).
- **Lose:** own ship sinks, a civilian vessel is hit, the target escapes more than 150 NM from its start point, or time runs out on a sink mission.
- Time warnings arrive at 5, 2 and 1 minutes remaining: on a sink mission as a deadline, on a survive (convoy) mission as the time until the convoy is safe.
- When the mission ends, R restarts it with the same seed (an editor mission restarts itself) and M returns to the main menu. During a mission, Esc offers "Main menu (without saving)" next to save-and-exit.
- Score: 1000 per submarine sunk, 200 per unused torpedo, 500 for no civilian losses, up to 500 time bonus.

Stations

The ship is split into nine stations. Keys 1-9 select a station; pressing the number of the current station again cycles its pages.

1 Bridge	2 Sonar	3 Weapons
4 Damage	5 Operations	6 Radio
7 Engine	8 Helicopter	9 Electronic warfare

Each station shows only what its sensors and operators know. Sonar contacts are noisy bearings until a ping, TMA, buoy or cross-fix supplies range. No station shows "the truth".

F2 hands the current station to the autcrew; F3 shows which stations run automatically. Use it when you want to concentrate on one or two stations. Shift+F2 switches the crew assist: the AI mans every station of both units that nobody holds, and the station on screen stays yours. A mission started from the multiplayer lobby has it on when a browser takes part or the uConsole is host only; started there alone it is a solo game with the assist off.

Controls

The game runs at 1280x720 and is designed for the uConsole keyboard and trackball. The trackball acts as a joystick: horizontal steers on the Bridge, vertical steps the station's main selection elsewhere. The uConsole can also be played entirely with the mouse (or the trackball's buttons): a click on a key in a station's key bar presses that key (holding the button holds the key, for example for steering or the telegraph), the numbered tabs in the top bar switch stations, a click on the course, speed or depth dial orders that value, and a numeric entry shows a keypad. Menu rows, dialog rows, save slots and the hints under them are clickable too; the wheel moves through menus and scrolls the help, and a right click cancels like Esc in menus, dialogs, entries and at the mission end. On charts the wheel zooms, dragging pans and a click pins a tooltip. A click does exactly what its key does, with the same checks.

Global keys (all stations):

Key	Action
Tab / Shift+Tab	Next / previous station
1 / 2 / 3 / 4	Bridge / Sonar / Weapons / Damage
5 / 6 / 7 / 8	Operations / Radio / Engine / Helicopter
9	Electronic warfare / ESM
Active station number	Press again to advance that station's page
Arrow keys	Station-specific selection or adjustment
+ / -	Telegraph (available at every station)
F1 / ?	Help (this display)
F2	Toggle Autocrew for the current station
Shift+F2	Crew assist: the AI mans every station nobody holds, on both units
F3	Open Autocrew overview
0	Weather & sonar analysis panel
F8	Tactical Unit Analyzer (read-only catalog)
F4	Simulation log view (live; requires simlog option; M: map of all contacts)
F9	Open local Commander LAN administration
F10	Options: language, fullscreen, audio, large text, tooltips, frame rate
F11	Event log and full telemetry overlay (station stays live)
N	Nations & units; in sonar: notch filter
S / L	Save / load (slots 1-5)
Alt+Enter	Fullscreen (all stations)
Q / E or mouse wheel	Chart zoom only on bridge, weapons, and helicopter
Drag	Pan chart (bridge, weapons, and helicopter)
K	Camera follow only on visible charts (drag disables it)
Left click	Press the clicked key, tab, dial or row
Right click	Cancel like Esc in menus and entries
P	Plot mode on the Bridge/Weapons/Helo map and OPZ chart: marks, rulers, bearing lines, circles, DR lines (shared by all stations, saved)
M R B C D · Enter · Bksp	In plot mode: choose tool, place point with Enter or click (arrows move the cursor, Shift faster), delete the nearest object (Shift: all)
Esc	Cancel input or open exit dialog
R / M	After the mission ends: restart with the same seed / main menu
D	After the mission ends: debrief with the truth beside what the crew knew

In the Remote Crew browser (Commander, F9) stations are operated with buttons; the keyboard helps with navigation:

Key	Action
1-9	Open one of your own stations (submarine crew: 1-7)
[/]	Previous / next own station
?	Open guide and station help
Arrow keys	Move within focused tab bar, track list or chart
Home / End	First / last entry of the focused list
+ / -	Zoom focused chart; Home fits the view
Arrow keys (chart)	Pan focused chart
Mouse over map	Details of the track, own ship, asset, wreck or chart position under the cursor
0	Open or close the weather & sonar analysis
Plot tool + click	Draw on the shared plot: pick a tool above the map, click once (mark, bearing line) or twice (ruler, circle, DR line)
, / .	Collapse or expand the contact list (,) or the station panel (.)
L	Open or close the operational log
Esc	Close the guide, lookout or contact library and return to the station

The bottom status ticker shows the newest event and key telemetry; F11 opens the full event log and telemetry over the station without stopping it or taking its keys. F1 (or ?) opens the help overlay at any time. It has four categories: global keys, the current station (keys and standard procedure), sensors and tactics, and this manual. Menus and dialogs over a running mission (help, options, save/load, quit, nations, F9, mission end) show the start screen's night scene behind a console panel instead of the station; the mission keeps running behind them.

Underwater acoustics in five minutes

- **Passive sonar gives bearing only.** Every contact starts as a line of bearing. Range comes from active ping, TMA, sonobuoys or a cross-fix.
- **Detection is signal against noise.** The passive sonar equation $SE = SL - TL - NL + DI - DT$ decides: the target's source level (louder = farther), transmission loss (spreading, absorption, layer and path losses), noise (own self noise plus wind, rain and nearby shipping), and the array gain. A contact is detected at $SE \geq 0$ dB. Wind and rain matter most when you run slow and quiet; at high own speed your own noise dominates.
- **Own speed is own noise.** Self-noise rises from 4 kn to 31 kn, the frigate's top speed. Above 15 kn the propellers cavitate and passive range drops to about a third.
- **The layer (thermocline) bends sound.** Passive propagation is ray traced through the real sound-speed profile: above the layer a surface duct carries sound far, below it lies a shadow zone a few miles wide. In deep water the shadow is strong; in water of a few hundred metres, bottom bounce and multipath fill it beyond about 10 NM, so hiding below the layer mainly works close in. A ping into the shadow zone reaches only 35 % of its range.
- **Seabed and surface matter.** Rock and gravel reflect sound well, silt and mud absorb it; a rough sea scatters high frequencies. **Convergence zones** appear only where the water is deep enough for rays to turn back up.
- **Baffles:** own-ship noise is a soft 70 degree lobe astern of the hull array (and along the cable of the towed array). It masks, it does not blank.

```

surface
~~~~~
frigate HMS )))          direct path
----- layer / thermocline ----- -6.5 dB across
      TAS below layer )))  submarine
                              (shadow zone for a shallow sensor)
_____ seabed

```

Navigation conventions

- Distances in nautical miles (NM), speed in knots (kn), depth in metres, frequency in Hz.
- Courses and bearings are true degrees: 000 north, clockwise. Charts are north-up.
- The game always runs in real time: one real second is one simulated second. There is no time acceleration and no pause; menus, help, options, save/load and losing window focus do not stop the simulation either.
- The ship turns at up to 0.8 degrees per second; speed changes take minutes. Plan manoeuvres early.

Your first patrol

1. Main menu: pick scenario 1 (Patrol) with 1 and start with Enter.
2. Bridge (1): press 1 again for the mission page, read objective and time limit.
3. Engine (7): select SLOW or 6-8 kn. Sonar (2): stream the towed array with Y.
4. Sonar BROADBAND page: look for a bright vertical trace; select it with the arrow keys and press Enter to follow it.
5. Classify with C, enable TMA with T, then turn the ship 30-60 degrees on the Bridge and hold the new leg for a few minutes.
6. When TMA or a ping gives range: release the contact to Operations (G), make it the target (M).
7. Weapons (3): set torpedo depth to the pinged target depth, fire with T.
8. Watch the sonar for an incoming torpedo; if one appears, go FLANK, turn away and stream the Nixie (V at Weapons).

Main menu, saving and options

- **First launch:** when no `~/.u-jagd/settings.json` exists yet, a welcome page follows the start screen: "What do you want to play?" 1 Frigate (the Training menu with lesson 1 selected), 2 Submarine (the Training menu with lesson 5, the first submarine lesson, selected), 3 Remote Crew (opens the multiplayer lobby; Esc there leads to the main menu), 4 or Esc main menu. Arrow keys and Enter choose as well. Whatever you pick, the page is remembered in the settings and not shown again.
- Menu: 1-7 scenario (4 = random with custom difficulty, 5 to 7 = submarine missions, see the reference chapter), W world mode, R new seed, F fullscreen, Enter start.
- **Training** (main menu): six guided lessons, each a short mission with a hint banner that waits for you: 1 listen and take bearings, 2 target motion analysis, 3 torpedo attack, 4 helicopter and sonobuoys; on the submarine (the uConsole plays the submarine for these two): 5 listen and hide below the layer (hear the frigate, classify it as a warship, measure the layer with a BT and dive below it), 6 shake off a hunting frigate (read the Threat page, evade with I, go quiet and deeper than 100 m until no ping has come for two minutes; the frigate pings every 45 s until you evade, then only while its pings still find you, and never fires). In lessons 1, 2 and 4 the submarine is neutral and never attacks; lesson 3 is a real attack that ends when the submarine sinks. The other lessons end as won after their last step. R at the end runs the lesson again. A saved lesson restarts its hints at step 1 after loading and passes the steps that are already done. After a lesson the uConsole keeps the side it played.
- **Campaign** (main menu): six linked missions in the sea area chosen in the menu (world W and seed; every mission keeps the same sector). Carried over are the torpedoes left (at least 2, at most 10), compartments still damaged, a lost helicopter and your standing with HQ (0-100, start 50: +15 for a

win, -20 for a loss, -10 for a civilian loss, +/-3 per task done or failed). After each mission the ship calls at port: 1 full refit (4 to 8 torpedoes by standing, all repairs, a new helicopter, standing -5) or 2 quick turnaround (half that restock, damage stays aboard, standing +3); Enter sails. The campaign ends when the ship is lost, standing falls below 10, or the sixth mission is done. It is kept in `~/ .u-jagd/campaign.json`, apart from the save slots: a slot saved during a campaign mission loads as a plain mission; to count, the mission is sailed again from the campaign screen. N starts a new campaign (twice while one is running).

- **Submarine campaign** (main menu, Campaign, Tab): five linked submarine missions in the same sea area, played on the submarine side: reconnaissance, breakthrough, convoy attack, breakthrough, convoy attack. Carried over are the submarine's torpedoes, its hull damage (at most 60 %) and your standing with submarine command (0-100, start 50: +15 for a mission won, -20 for one lost). After each mission the submarine calls at its base: 1 full refit (full torpedo load, hull repaired, standing -5) or 2 quick turnaround (half of 4 to 8 torpedoes by standing added, the hull damage stays, standing +3); Enter puts to sea. It ends when the submarine is lost, standing falls below 10, or the fifth mission is done, and is kept in `~/ .u-jagd/boat_campaign.json` beside the frigate's campaign.
- **Logbook** (main menu): every finished mission (never a lesson) for the side the uConsole played, with date, mission, realism level, result, score and minutes; the best score per mission and five awards per side: first victory, one shot one kill (the enemy sunk with a single weapon), unscathed (no damage), never fired at, and realist (a victory on the Realistic level). The frigate files its mission score; the submarine counts its outcome (sinking the frigate 1500, sinking the convoy 1200, breakthrough or report 1000, escape 800, surviving 600) plus up to 500 for an undamaged boat and 100 per torpedo left, times the level's factor. Left/Right switch frigate and submarine, Esc back. The end panel names the score, a new best and new awards. The logbook is `~/ .u-jagd/logbook.json` (the newest 200 missions), never part of a save.
- **Report a bug** (main menu): writes `~/ .u-jagd/bug-report.txt` (version, platform and the newest lines of `~/ .u-jagd/crash.log`, with your user name removed from paths) and shows a QR code that opens a new GitHub issue on a phone with version and platform filled in; attach the file there. Enter opens the issue with the log in a browser if the device has one, Esc goes back. After a crashed start the main menu selects this entry and says so. Nothing is sent until you submit the issue with your own GitHub account. The browser settings menu has the same link.
- **Update notice** (start screen and main menu): new releases are never installed on their own. At launch the game asks GitHub once whether a newer release exists; if so, the start screen (top right) and the main menu (left of the entries) show its version, its changelog entry in the game language and, when saved games of this version (the autosave too) will not load in the new release because the save format changed, a warning. U or a click on **Update now** installs it: on the uConsole the game closes, the launcher's small window shows the download and check and the new version starts; the Windows program downloads it in the background (progress on the button), checks its size and SHA-256 digest, closes, replaces itself and starts the new version; any other installation opens the release page. Offline, or with `U_JAGD_NO_UPDATE=1`, no notice appears.
- S / L: save / load (slots 1-5). Saves are exact and deterministic: a loaded game continues identically.
- **Autosave**: a running mission is saved every 5 minutes and when you quit or leave it for the main menu, to `~/ .u-jagd/autosave.json` beside the five slots. The main menu then starts with **Continue mission**, which resumes it exactly; after a crash it holds the last 5-minute save. A mission that ends (won, lost or ship sunk) and any new mission delete the autosave. The web host (- -web-host) does not autosave.
- F10: options - language, fullscreen, audio, large text, tooltips, frame rate (30 or 60 FPS; 30 saves CPU on the uConsole and is the default), event log / telemetry as status ticker (default, more room for the station) or docked band, and the **realism level** for the next mission: **Beginner** (operator assistance on: automatic line labels, blade-rate and catalogue/emitter candidates; the computer opponent attacks less eagerly, waits for a better firing solution and, as the frigate, classifies and launches its helicopter 1.5 times slower; score 75 %), **Standard** (default: raw data and manual analysis, the calibrated opponent; score 100 %) or **Realistic** (no assistance; the opponent attacks more eagerly, fires on a rougher solution and reacts 30 % faster as the frigate; score 125 %). The level only tunes the computer opponent, never a human on the other side, and a running mission keeps the level it started with (the row then says "from the next mission"). The end panel shows the level with its score

factor; saves keep it. Page 2 (PgDn/Tab): which side the uConsole plays, frigate (default) or hostile submarine; only in the main menu, never saved. A new game asks for it first anyway. Page 2 also holds **anti-aliased chart lines** (off by default; smooths bearing lines, coast and plot at some CPU cost on the uConsole). And **spoken crew reports** (off by default): the crew says torpedo in the water, new contact with bearing, breaking-up noises, torpedo away, hit, action stations, patrol aircraft on station and the mission result aloud, bearings digit by digit. The uConsole speaks through an installed `espeak-ng` (`sudo apt install espeak-ng`) and stays silent without it; Remote Crew browsers have their own switch under Settings (the browser's speech synthesis, in the browser's language). With the uConsole on the submarine the submarine's crew reports instead (see the reference chapter).

- **Multiplayer** (main menu): the lobby where the crew meets before a mission. It starts Remote Crew in crew mode by itself and shows the QR code, address and join code. On the uConsole's own hotspot it shows two numbered steps side by side: **1** the Wi-Fi QR code with the Wi-Fi name and password (join the hotspot), **2** the page QR code with the address and join code (open the crew page). Browsers pair, choose their unit and stations and press **Ready**; they see the mission, what the uConsole plays and every crewmate with their stations and ready tick. On the uConsole, Up/Down choose a row and Left/Right change it: the mission, the unit the uConsole plays and the station it shows, or **none, host only**: then the uConsole plays no station, the browsers can take every one and the AI crews the rest. **Start the mission for everyone** starts a five-second countdown that every browser sees, then the mission begins for all at once and the uConsole opens on its chosen station. If a crewmate with a station is not ready yet, the first Enter asks again and a second one starts anyway. Esc cancels a countdown, otherwise it leads back to the main menu while Remote Crew keeps running. When a mission started from the lobby ends, everyone returns to the lobby with their stations; the ready ticks start again from zero. Every mission started from the lobby with a browser taking part (or host only) has the crew assist on (Shift+F2); alone it starts as a solo game with the assist off. F9 opens the full Remote Crew settings from the lobby. Started with `--multiplayer`, the game opens the lobby straight after the start screen.
- F9: Commander / Remote Crew - lets browser clients on the LAN take stations. The page has one switch, **Multiplayer**: Enter turns it on on the first local network address, or, when the uConsole has no network, on its own hotspot (if the hotspot helper is installed; the uConsole installer `install.sh` sets it up when it can and otherwise prints a note). The hotspot keeps its Wi-Fi name and password from one start to the next, so a phone or PC that joined once reconnects by itself. On the hotspot the page shows the two steps together: **1** the Wi-Fi QR code with name and password, **2** the page QR code with the join code. **Crew** lists the players and their stations. **Advanced network settings** shows the network mode (LAN or hotspot), the address and the port for a manual choice; they change only while multiplayer is off. There, while multiplayer is off, **New hotspot password** makes a new hotspot password and keeps the name; every device then has to join again with the new Wi-Fi QR code. A free station is taken at once with all of its rights (including direct fire and live sonar audio where the station has them); a station a crewmate holds is requested, and the holder (who sees the request with Hand over / Keep station buttons) or the host can hand it over. A station always carries its full rights; the host can revoke a station, grant or withdraw the SimLog (roster key L) at any time, and can make up to two browsers read-only observers (roster key O): they watch any station of either unit without holding it, cannot command, and get the SimLog with a debrief timeline and JSON export. On a Windows PC the program `U-Jagd-Windows.exe` starts straight into the game with the same command-line options; there is no separate starter window, and **Multiplayer** in the main menu (or `--multiplayer`, see the README) opens the lobby with Remote Crew on. The crew pages open in the host's saved language (F10 options on the uConsole); the English/Deutsch button in the browser's status bar switches that browser alone. The crew page is built for Chrome or Chromium (also Edge) on a desktop PC; another browser shows a hint above the pairing code, and a page that cannot start there says so instead of loading forever. After a host update an open browser page reloads itself once, so it always runs the web client that matches the host.

Phone lookout and periscope

A phone can stand the watch as the frigate's bridge lookout or on the crewed submarine's periscope. F9 shows a second QR code, **Phone lookout**, for the address `https://<address>:<port+1>/lookout`. Scan it, accept the certificate warning once, choose the watch station, and type the pairing code shown next to it (the code is never in the QR code). Type it as shown, with or without the space and in any case; look-alikes such as O and 0, l, I and 1 or S and 5 are read by position. "Wrong pairing code" means exactly that and shows the code the game received; if the game refuses the address itself, the page says so.

- **Certificate:** the game makes its own certificate for its LAN address (kept in `~/ .u-jagd/tls/`, renewed when the address changes). The phone warns once because no authority signed it: on iPhone tap **Show Details**, then **visit this website**; on Android Chrome tap **Advanced**, then **Proceed**. Phones only hand the gyroscope and the microphone to such a secure page.
- **Looking around:** tap **Gyro** and turn the phone like binoculars; tilt it to look up or down. Without the gyroscope, swipe. **Ahead** looks at the bow again, **Zoom** cycles the magnification. On the periscope the phone trains the periscope itself, and **Range** takes a stadimeter range on what is in the crosshair.
- **Reporting:** tap **Report by voice** and say what you see, for example "Ship bearing 040, range 5 miles", "aircraft starboard 30" or "torpedo" (the line of sight then counts as the bearing). Categories: contact, ship, warship, merchant ship, aircraft, submarine, torpedo. Or tap the target in the picture and pick the category.
- **Confirmation:** a report counts only when the lookout really has something of that kind within 10° of the bearing (and, with a range, within 40 % or 1 NM of his estimate). Then it appears on the bridge as a lookout track and in the event log, and the crew browsers speak it. A report of nothing is refused, and the phone vibrates twice.

While a phone holds the bridge lookout, the lookout no longer reports ships, aircraft or torpedoes by himself: only what the player calls reaches the bridge (land is still reported automatically). On the submarine the periscope picture stays with the attack computer, and the crew's own "in sight" notices give way to the phone's reports. Speech recognition uses the phone browser's speech service (Chrome on Android, Safari on the iPhone with Siri and Dictation switched on; Firefox and the other iPhone browsers have none, tap the target there). When it fails the page names the reason. Nothing of the phone lookout is saved.

1 Bridge

Purpose

The Bridge conns the frigate: course, speed and position relative to coast, contacts and threats. Every sensor depends on how the ship is driven. Fast and straight is loud and blind; slow, steady legs with deliberate turns make sonar and TMA work.

Displays and instruments

The top bar shows the station, the mission, the clock, speed and course; the chart header shows only its scale (plus "follow" while K follows own ship). Page 1 (navigation) shows the chart and four panels; page 2 (press 1 again) shows the mission briefing; page 3 is the lookout scope. The chart water darkens with the clock in three steps (day, dusk within an hour of 05:30 and 19:30, night), and rain or a storm hatches the chart with dashed diagonals (a storm adds an amber border); both are display only, as on the browser chart. Options page 2 can anti-alias the chart and plot lines.

CHART (north up)	COURSE / RUDDER
own ship + wake	course 045 > 080
published tracks	rudder 15 R
bearing lines / fixes	SPEED / ACOUSTICS
helicopter, buoys	HALF 10.0 kn
	38% own noise
	TACTICAL PICTURE
	threats, sensors,
	assets, weather
footer: <- -> course Up/Down telegraph U/V direct	

- Course / rudder:** current course, ordered course (→), rudder angle in whole degrees and, only while turning, the turn radius. Below the numbers a rudder scale shows the angle from port (left) to starboard (right), and a compass dial shows the heading needle with the ordered course as a hollow amber mark.
- Speed / acoustics:** telegraph order, speed, own noise in percent and a CAVITATION warning above 15 kn. A speed dial from 0 to 31 kn shows the present speed as a needle and the ordered speed as a hollow amber mark.
- Tactical picture:** observed threats (a heard torpedo launch transient, HF seeker pulses or a submarine flooding a torpedo tube, a contact sonar classified as torpedo, or an air track flagged as a possible missile), sensor state (radar, TAS), assets (helicopter, buoys) and weather/day-night. Beside the weather lines a small picture in the start screen's look looks into the wind: the sky of the hour with sun, moon or stars, the clouds, rain, snow or fog and the sea running at the eye, with a wind rose (north up, the arrow blowing downwind) in its corner; the Remote Crew bridge shows the same picture.
- Chart:** synthetic chart depth and coastline, own ship, tracks published by the other stations. Q/E zoom in fixed steps (chart height 500, 250, 100, 50, 25, 10, 5, 2, 1 and 0.5 NM), the wheel zooms smoothly down to 0.5 NM; the grid gets finer as you zoom in (down to 0.1 NM). Drag pans, K follows own ship.

Bridge lookout reports

The bridge lookout (eye height 18 m, 7x50 binoculars) reports sightings in the event feed as AUSG lines, for example Bridge lookout: frigate (Admiral-Gorshkov-Fregatte) bearing 040°, 3.8 NM. The Remote Crew bridge lists the same reports under "Lookout reports". Bridge page 3 (the lookout scope) shows the sightings north up around the own ship at the range and bearing the lookout measured, coloured by kind (surface, submarine, aircraft, torpedo) and labelled with what he made out, next to visibility, sea state, day/night and the latest reports; , and . change the scope radius (2 to 30 NM). Above the reports a horizon strip shows the binoculars toward the bow (90° field, true-bearing scale, the horizon moving with the sea, the light of the hour) with the outlines of the lookout's sightings at their measured bearing and range; it is the same renderer as the submarine's periscope. The picture has the start screen's look: by day a blue sky, at dusk a warm horizon, at night stars and the moon in its phase at its bearing with its glitter on the water; clouds, rain, snow and fog follow the weather station, and the silhouettes are drawn in steel with a lit rim, with lit windows at night. Once the lookout has made out the class of a ship, submarine or aircraft, it is drawn as its 3D model (the analyser's), turned by the angle on the bow he judges in 10° steps, as soon as it is at least 16 pixels long; before that, when stale or smaller it stays a flat silhouette. The model is the real type the eye sees (every ship, submarine and aircraft type has its own model, see the reference chapter), so the type can be told by sight; the lookout's report names only what he made out, and only the report goes to the OPZ. From dusk to dawn and in visibility under 2 NM neutral merchant ships and fishing vessels run their navigation lights (warships run darkened): white masthead lights over the forward 225° (two from 50 m length, the aft one higher, 6 NM), the green starboard or red port side light (3 NM), both when she heads straight at you, and the white stern light over the 135° astern (3 NM; under 50 m length one masthead light at 5 NM and the others at 2 NM), never beyond the visibility. Vessels at work add their all-round lights: a trawler green over white (a masthead light only from 50 m), a pilot vessel white over red instead of masthead lights, a survey ship, cable layer or research ship restricted in her ability to manoeuvre red, white, red, and a mine clearance vessel three green; a tug without a tow shows ordinary lights. Civil aircraft show the red left and green right wingtip light and the white tail light (3 NM) and their flashing red anti-collision beacons and white strobes (10 NM); military aircraft fly dark; the silhouette then points its bow the way the lights show, and a lit ship is sighted by its lights even where the dark hull is not (its class still needs the silhouette). The lookout calls out the lights he sees with his reading of them, for example Bridge lookout: lights bearing 040°, 2.8 NM: two masthead lights, red side light; showing her port side: both side lights mean she is heading for us (called aloud), green alone her starboard side, red alone her port side, the stern light alone that she is going away, and the all-round lights her work (fishing, pilot on duty, restricted in ability to manoeuvre, clearing mines; flashing lights an aircraft). He calls a contact's lights again only when what they tell changes, at most every 2 minutes; the Remote Crew bridge lists these calls with the other reports. Aircraft stand at their true elevation above the horizon, worked out from their height and range less the curve of the Earth, and hang in the still sky behind the clouds instead of riding the swell; an aircraft high and close is above the field until the binoculars are tilted up. The Remote Crew bridge shows the same binoculars as a card ("Lookout binoculars"), trained in that browser only with the arrow buttons (2°, 10°) and "Bow". B raises the binoculars large over the chart: a 16° field the operator trains with , and . (Shift: 20° steps) or by clicking the all-round panorama below it, which marks every sighting at its measured bearing with the bow in the middle; the sightings are listed underneath, nearest the line of sight first. The charted coast stands on the horizon of the strip and the binoculars as far as the lookout can see land (at most 20 NM, fading into the haze), and the panorama marks it along its foot; the chart has no elevation, so the hills are an assumed 25 to 70 m. While the binoculars are up, ↑/↓ tilt them 2° (Shift: 10°, from 20° down to 45° up) instead of working the telegraph, Q/E zoom them (16°, 8° or 4° field) and Space switches the stabilizer, which takes out all but an eighth of the ship's motion; the line under the picture shows tilt and field, and the Remote Crew card has the same buttons for its own browser. The sea follows the wind: looking into it the crests come at you in long rows, looking down-sea their backs run away, across it short crests run sideways, and the ship pitches in a head or following sea and rolls in a beam sea. The waves fill the whole sea in perspective, small and close together out to a clean horizon, longer and higher toward the eye, and every row moves with the swell. The ship's own way shows in the water: underway the rows stream toward the eye looking ahead, away from it looking astern and from bow to stern looking abeam, faster with more speed; astern the wake runs as a band of smoother, lighter water with foam between the two arms of the Kelvin wave

out toward the horizon, and ahead the bow wave throws its spray up into the lower edge of the picture (tilt the binoculars down to see the white water curl out from the stem). B again returns to the chart; the binoculars are display only and are not saved. A contact is reported in up to three steps as it closes, each step once:

- **Sighted:** only the kind of object is clear (vessel, aircraft, small object on the surface).
- **Class:** the silhouette shows the class, for example merchant ship, warship, aircraft carrier, fishing vessel, speedboat, surfaced submarine, airliner or military aircraft.
- **Type:** close in the lookout names the type: cargo ship, tanker, passenger ship, tug, frigate, destroyer, corvette or combat aircraft; warships and military aircraft also with their class name. Merchant ships and airliners are identified by name, AIS or transponder, not by eye, so the lookout never reports their name or airliner type.

Class and type need a finer resolved silhouette than the sighting (Johnson criteria): by clear day a tanker is classed at about 7 NM and a frigate identified at about 4 NM, a speedboat is classed only inside 3 NM, and at night the type is made out only within a few cables. Fog, rain and sea state shorten every step. The lookout also calls "land in sight" with the bearing of the nearest coast, and a torpedo wake with a banner. The class is held while the lookout keeps the contact. It appears in the chart and OPZ tooltips as "Lookout: ..." and is an observation only: it never sets the OPZ classification or the affiliation.

Autopilot route

The helm can follow a route of up to 8 waypoints. On the navigation page a right click on the chart adds a waypoint; W starts a search pattern from the ship's position and course (first a zigzag of 3 NM legs 45° either side of the course, pressed again an expanding square of 1, 1, 2, 2, 3, 3, 4, 4 NM legs turning right, a third time the route is cleared), and Backspace clears it. The chart draws the route as an amber line with numbered waypoints and the Course panel shows the next one with its distance. The autopilot sets only the ordered course; speed stays with the telegraph. A waypoint counts as reached within 0.3 NM, then the helm steers for the next one; after the last one the ship holds its course. Any helm order (←/→, U, the trackball or a course from Remote Crew) takes over and switches the route off; with the Bridge out of action no route can be set and an active one is not steered. The autopilot plans on the chart: charted depth and charted rocks and wrecks against the hull's minimum depth (draft plus keel reserve) and another 2 m margin, sampled every 0.1 NM along each leg and 0.1 NM either side. A new waypoint or a search pattern whose leg crosses shoal water or land gets detour points: first a stand-off point square to the leg beside the first shoal point (nearest first, port before starboard); when that does not clear it, a path search on the chart (a grid of at most 96 cells a side over a box 4, then 30, then 80 NM around the leg, straightened into as few turning points as the clear legs allow) finds the way through a channel or round a bay or a long coast. The route may then hold up to 16 points; when no way is found the feed warns which leg to steer by hand. While the route runs the autopilot looks ahead on the leg (two minutes at the present speed, at least 0.5 NM) once a second: shoal water there gets a detour to the current waypoint, or the route is switched off and the ship turned back on the reciprocal course. It plans on the chart, not the live tide. The route is saved. The Remote Crew bridge has an "Autopilot route" card: "Set waypoints on chart" makes a click on open chart add a waypoint, and buttons start the zigzag or the expanding square or clear the route.

Keys

Key	Action
<- / ->	Rudder: change target course
Up / Down	Telegraph up / down
U	Enter target course directly (000-359)
V	Enter target speed directly (0-31 kn)
+ / -	Telegraph: engine order (ASTERN-STOP-SLOW-HALF-FULL-FLANK)
Chart	Mouse wheel: zoom, mouse drag: pan
Q / E	Zoom chart in steps, 500 to 0.5 NM
K	Camera follow on/off
, / .	Lookout page: scope radius smaller / larger
B	Lookout page: binoculars over the chart on/off (, / . train them)
↑/↓ · Q/E · Space	Binoculars up: ↑/↓ tilt 2° (Shift: 10°) instead of the telegraph, Q/E zoom (16°, 8°, 4° field), Space stabilizer
G	Action stations on/off
W	Autopilot: zigzag search, expanding square, off
Right click	Add an autopilot waypoint on the chart
Backspace	Clear the autopilot route
Ctrl+B	Clear baffles: 60° to starboard for two minutes, then back (the hull sonar is deaf 30° either side of the stern)

The trackball steers the rudder while the Bridge is selected. U and V open direct numeric entry; the simulation keeps running while you type. Enter confirms, Esc cancels.

Standard procedure

1. Check mission page (1 twice): objective, time limit, area.
2. Transit at FULL, then slow to SLOW or 4-6 kn before the search area.
3. Search: steady legs of 3-5 min, then turn 30-60 degrees so TMA gains range.
4. Keep clear of the coast and shallow water; watch the chart depth.
5. Torpedo inbound: FLANK, turn away beam-on, order Nixie at Weapons.

Combat situation:

1. Torpedo reported: FLANK immediately, turn to put the torpedo bearing astern or on the beam.
2. Order the Nixie at Weapons (V); keep turning so the torpedo sees the decoy first.
3. Once clear, reduce speed below 15 kn so sonar can reacquire; never above 20 kn with the towed array out.

Pro tips

- TMA needs a real change of own velocity. A 30-60 degree turn followed by a steady leg of several minutes gives the best range estimate. Turning on the spot does not help.
- The turn rate grows with speed (about 0.75 deg/s at 10 kn, 1.2 at 16 kn, 1.9 at 25 kn, 2.3 at 31 kn), so the turning circle stays near 0.4 NM. A stopped ship cannot turn. Speed changes take minutes:

about 90 s to 90 % of FULL, and a stop from FULL uses reverse propeller pitch and takes about 90 s. Start evasive turns early.

- In a hard turn at speed the ship heels outward a few degrees; in heavy seas the fin stabilizers damp the roll, but only with steerage way.
- In shallow water the hull squats: at 25 kn the draft grows by up to 3 m, at 31 kn by up to 4.6 m, when the water is less than about five draughts deep. Slow down in shoal water.
- Sprint-and-drift: sprint at FULL to a new position, then slow to 4-6 kn and listen.
- Heavy flooding on one side gives a list and a steady yaw pull; correct with rudder.
- The ship cannot run aground onto land; it is pushed back, but shallow water limits the helicopter dipping depth (10 m bottom clearance).
- Water depth follows the tide (semi-diurnal, about 12.4 h, up to a few metres in shallow water). A passage that is safe at high water can ground the hull at low water; the HQ weather bulletin reports the current tide at the ship.
- Wind pushes the surface water: about 3 % of the wind speed, 20 degrees to the right of downwind, on top of the steady ocean current.

Not modelled

- No time acceleration and no pause. The autopilot steers only course, never speed; its detours use only the chart (no live tide, no other ships); its path search sees the chart in cells of at least 0.2 NM, so a channel narrower than about half a mile, or a way round farther than 80 NM from the leg, is not found.
- No automatic torpedo identification: the alarm rests only on heard intercepts or the sonar operator's classification; a torpedo running silent outside seeker range can arrive unannounced. The lookout calls out only a visible wake.
- The lookout never reads a ship's name or flag and does not report day shapes; he reads lights only as listed above, never a course or speed from them. No ship in the game anchors, tows, trawls at trawling speed or is not under command, so anchor lights, towing lights and the not-under-command lights never show.

2 Sonar

Purpose

The sonar room is the main ASW sensor. It listens passively on the hull-mounted sonar (HMS), the towed array (TAS) and the variable-depth sonar (VDS), analyses signatures on LOFAR and DEMON, estimates target motion with TMA, measures the sound profile and, when ordered, transmits an active ping. It classifies contacts and releases them to Operations and Weapons.

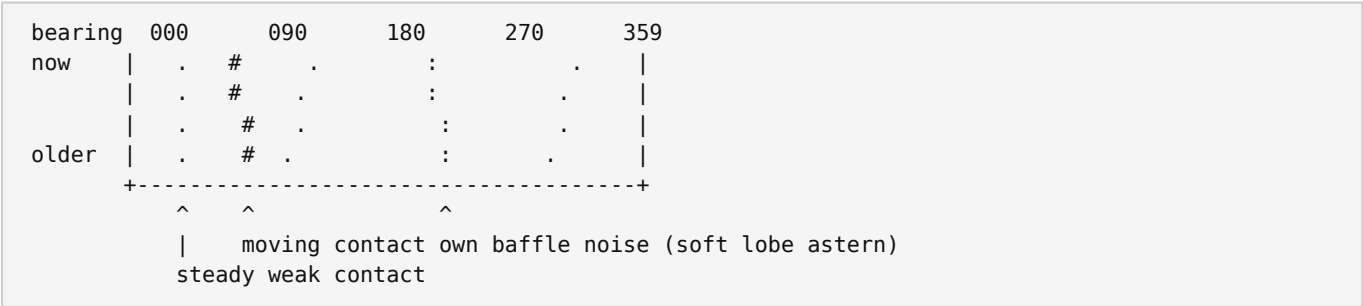
Displays and instruments

The station has six pages. PgUp/PgDn (or 2 again) cycles them. The listening post on the right shows the evidence source and age, the array, the listening bearing and the ping readiness; a towed or variable-depth array that is moving or not yet ready appears on every page in amber with payout, stability and PAUSE when handling is outside its limits. The single key row at the bottom shows the page's four main keys with their values; every other key is listed under F1. On BROADBAND the amber line is the listening bearing and the grey lines the beam limits; dark means quiet, turquoise loud.

Page	Shows	Use it for
BROADBAND	Bearing-time waterfall	Detecting contacts and following their bearing
LOFAR	Frequency-time waterfall, 0-300 Hz	Tonal lines: machinery, shaft, blade signatures
DEMON	Modulation spectrum of the listening beam	Blade rate and shaft-rate hypotheses
TMA	Solution for the focused contact	Range, course and speed estimates
UMWELT / FUSION	Sound profile, layer, CZ, array comparison	Choosing TAS depth, spotting ghost contacts
ACTIVE	Stored echoes with age and error	Range and depth from pings

The listening console below the detail rows works like a control desk: lamps show ping ready (amber while a ping is out), audio and peak hold, and a north-up bearing rose shows the listening beam (amber), the baffles astern (red sector), own course and every published contact bearing. The contact list marks each contact with a lamp and a signal-to-noise bar. The Remote Crew browser shows the same rose beside its waterfalls.

BROADBAND waterfall



Newest data is at the top. A straight vertical trace is a contact on a steady bearing; a slanting trace shows bearing rate. The history covers 20 s of fine data plus 4 minutes of long history (Shift+H selects 25/50/100 %). Brightness is relative received level, not range.

LOFAR

The x axis is frequency (0-300 Hz), time runs downwards. Bins are 1 Hz below 40 Hz, 2 Hz up to 100 Hz and 5 Hz above. A new line is added every 0.25 s; 80 lines are kept.

- Steady vertical lines are **tonals** (narrowband): generators, pumps, shaft lines. Several lines at integer multiples of one frequency are a harmonic family: put the white cursor on a line with Z/X (Shift: 10 Hz steps) and press K to mark it as fundamental; amber guides then show 2f, 3f and so on. K on the same frequency clears it.
- Own ship produces a shaft line at about $10 + 1.9 \times \text{own speed Hz}$. N notches it out.
- Space holds peaks so faint tonals stand out.
- The spectrum strip above the waterfall prints the frequency over every prominent line (interpolated between bins; with Space the held envelope). Where values would overlap, the stronger line keeps its label. The Remote Crew browser labels its spectra the same way.
- F selects the analysed band: FULL 0-300, LOW 4-80, SHAFT 8-55, MID 20-120 Hz. Ctrl+Z / Ctrl+X set the low / high band edge at the cursor for any band-, low- or high-pass; Shift+N puts an extra notch on the cursor frequency.
- Q selects the integration time: 2 s (the receiver's own FFT), 8, 16 or 64 s. Longer integration averages successive spectra so a weak steady tonal rises out of the noise, but a moving line smears. Shift+Q opens the vernier: 20 Hz around the cursor at the native 0.5 Hz resolution.
- The detail rail reads the level at the cursor. The strip labels lines automatically only at the Beginner realism level (F10).

DEMON

DEMON demodulates the broadband noise envelope of the listening beam. Propeller cavitation is modulated at the **blade rate** = shaft rate x number of blades.

```
level
|      |
|  |  |      blade rate 12.5 Hz
|  |  |      -> 5 blades => shaft 2.5 Hz = 150 RPM
|  |  |
+---+---+---+---+ Hz
  2.5 12.5 25
shaft blade 2nd harmonic
```

The display shows measured modulation, not identity. After changing the bearing listen for at least a few seconds before judging. Count blades yourself: move the cursor (Z/X, 0.5 Hz) onto the shaft line and press K, then onto the blade line and press K again; the rail shows blades = blade rate / shaft rate (with the deviation from a whole number) and the shaft RPM. A third K clears both marks. The **class library** under the marks lists the three catalogue classes that fit your marks best: shaft RPM against the class's RPM range, the blade count against its blade counts and your LOFAR fundamental (K on the LOFAR page) against its tonal band and machinery lines, each best in the middle of the range. With fewer than two marks it is only a hint. The contact analyser (F8) then lists the whole catalogue best fit first with the fit in percent. The library grades your marks, never the contact itself, and the classification stays your decision; the submarine's sonar room has the same library. At the Beginner realism level (F10) the sonar also labels modulation lines, proposes RPM for 3-7 blades and ranks catalogue candidates.

TMA, environment and active

- **TMA** is yours. The page plots the bearings of the selected contact over time. Build a hypothesis: Z/X course (Shift: 1 degree), Ctrl+Z/Ctrl+X speed, Q/Shift+Q range on the newest bearing (Ctrl: 0.2 NM). The amber curve shows the bearings that hypothesis predicts, the dots at the foot the residuals (measured minus predicted). Good hypotheses leave residuals scattered around zero; a wrong course, speed or range leaves a trend. The rail shows the residual RMS, the systematic trend after averaging,

the fit and the observability. Range is only observable after an own course change (at least 6 degrees, better 30-60): without one, K refuses. K accepts the hypothesis as the contact's TMA fix; it is dead-reckoned on its course and speed and ages out after 120 s, so refine and re-accept as bearings come in. Noisy bearings give a large range uncertainty even when the fit is good. TMA does not estimate depth. At the Beginner realism level (F10) an automatic solver proposal (it needs at least 4 bearings over 180 s) is drawn as a thin line and Shift+K copies it into the hypothesis. Sonobuoy bearings enter the track with the buoy as observer. Shift+T switches the TMA method: HYPOTHESIS (above), EKELUND (the range from the bearing rates of two own legs about one course change of 30 degrees or more, 90 s each; Shift+K copies it into the hypothesis, K then accepts as usual) or DOT STACK (residual rows at 0.6, 1.0 and 1.6 times the hypothesis range: the flat row is the range the bearings support).

- **UMWELT / FUSION** shows the bathythermograph (E, 60 s cooldown): measured layer depth, sound-speed profile and convergence-zone bands, plus the HMS/TAS comparison. Bearings within 5 degrees confirm each other; 9 degrees or more apart are flagged as a possible ghost contact. The layer is not fixed: afternoon sun makes it shallower (about 8 m), strong wind mixes it deeper over hours, and internal waves move it a few metres. Repeat the BT after a few hours or a weather change. The measured profile is the real temperature-driven sound speed (Mackenzie equation), so it drops below the layer. Point the mouse at the profile to read the exact depth, the sound speed there and whether that depth lies above or below the layer (uConsole and web). In the web client the plot also labels the layer, the seabed and the sound-speed minimum.
- **ACTIVE** lists echoes of the last 120 s: bearing, range and depth (+/-12 m). A ping fix ages out after 120 s. W selects the pulse: **CW** (1 s tone) gives coarse range (about 0.1-0.3 NM) but its Doppler separates a moving target from seabed reverberation; **LFM** (100 Hz sweep) measures range to a few metres and gains 20 dB against noise, but a slow or stationary target stays inside the reverberation. The echo strength depends on the target's aspect (broadside about 15 dB stronger than bow-on) and size. Rocky ground reverberates far more than mud; charted wrecks return real echoes that no contact owns ("unassociated echo").
- **Hearing the echo:** every return is audible when it arrives, after its real two-way travel time (about 2.5 s per nautical mile of range), for the ship's sonar and the helicopter's dipping sonar alike. A CW echo is a soft tone on the carrier, an LFM echo a short sweep; the pulse is the one the ping was sent with. A strong echo stands out clearly, a faint one barely rises out of the reverberation hiss. In the Remote Crew browser the echo plays with the general sound (loud or faint). A submarine's active ping is heard the other way round, after its one-way travel time (about 1.2 s per nautical mile): only then does the frigate sound its low warning tone, play the ping itself and log it with the bearing measured by ear (whole degrees, about $\pm 2^\circ$ off); a ping never gives its range. **Directional hearing:** with stereo sound (headphones), echoes, detonations and a submarine's ping come from the bearing they were heard on: left for port, right for starboard of the ship's head, centred dead ahead and astern (ahead and astern sound the same). The ship's own ping, launches and gunfire stay centred. The uConsole and the Remote Crew browser place them alike.

Hull sonar, towed array and VDS

	HMS (hull)	TAS (towed)	VDS (variable depth)
Passive range	1.0 x base	1.4 x base, minus 3 % per knot	1.15 x base
Beam width	12 degrees	6 degrees	8 degrees
Bearing error	+/-6 degrees	+/-2 degrees, left/right ambiguous	+/-4 degrees, unambiguous
Self-noise	full	35 % of hull	60 % of hull
Ping range	1.0 x	0.8 x	1.1 x, from the body's depth
Handling	always ready	stream 360 s, recover 480 s, only at 3-12 kn, 30 s settle	lower 120 s, recover 120 s, only at 3-15 kn and sea state 5 or less, 20 s settle

- TAS depth 20-260 m (U/V in 10 m steps), limited to 260 m minus 4 m per knot of own speed. At 30 m or deeper and in the same layer as the target it gains another 25 %.
- Above 20 kn with any cable out the array suffers a permanent FAULT.
- The array heading lags the ship by about 45 s after a turn; its bearings are less reliable while it swings.
- **Baffles:** the hull array (HMS) is deaf 30° either side of the own stern, so a boat right astern is heard only by the towed array or the VDS. The BROADBAND waterfall marks the baffle edges with dotted lines. Clear them from the Bridge with Ctrl+B (two minutes 60° to starboard, then back) or with any turn of your own. The same holds for the enemy: a submarine's hull sonar is deaf astern too, and an AI boat that finds itself close in the frigate's baffles after a ping follows the frigate there instead of running.
- VDS (Shift+Y lowers or recovers it, Shift+B selects it): a body on a short cable, 20-300 m deep and limited to 300 m minus 8 m per knot. U/V move the depth of whichever array is selected. At 30 m or deeper in the target's layer it gains the same 25 % as the TAS. It is unambiguous, so a VDS bearing resolves the TAS side just like the hull sonar. Pinging on the VDS transmits from the body: below the layer the shadow-zone loss hits shallow targets instead of deep ones. Lowering and recovery pause outside 3-15 kn or above sea state 5; above 24 kn with the body out it is lost (FAULT).

HMS	TAS
noise lobe astern	noise lobe along cable
^	^
\ ship /	ship ===== cable === array
\ /	(lobe points to ship)
~~~~  ~~~~ 70 deg soft lobe	

## Keys

Key	Action
Shift+A	Transmit active ping (cooldown; reveals position!)
Shift+B	Switch receiver array: HMS, TAS, VDS
Y	Deploy / retrieve TAS (only at 3-12 kn)
Shift+Y	Lower / recover VDS (3-15 kn, sea state up to 5)
Page Up / Down	Broadband / LOFAR / DEMON / TMA / Environment / ACTIVE
2	Press 2 again to advance the sonar page
E	Bathymograph: measure local sound profile
W	Active pulse CW / LFM
U / V	Raise / lower the selected array's target depth (TAS or VDS) by 10 m
R	Direct listening bearing: 000 to 359.9 degrees true
<- / ->	Bearing +/-0.5 degrees; Shift: 5, Ctrl: 0.1
Up / Down	Select contact for TMA and classification
Enter	Follow measured contact bearing / hold manually
J   , / .	Receiver audio on/off   lower/raise volume
A / B / H	Direct broadband / filtered / heterodyne audition
D	Toggle broadband/filtered audition
I / 0	Lower / raise gain (3 dB)
Shift+I / Shift+0	Lower / raise sonar display contrast
Ctrl+I / Ctrl+0	Lower / raise sonar display black level
Shift+C	Cycle green / amber / cyan phosphor palette
Shift+H	Cycle displayed history depth: 25 / 50 / 100 percent
F	Select frequency band: wide / low / medium
N	Notch filter against own propulsion
K	Mark line at cursor (LOFAR fundamental, DEMON shaft/blade)
Z / X	LOFAR/DEMON frequency cursor (Shift: 10 Hz)
Ctrl+Z / Ctrl+X	Band-pass low / high edge at the cursor
Q	Integration time 2 (FFT)/8/16/64 s
Shift+Q	LOFAR vernier: 20 Hz at native 0.5 Hz
Shift+N	Notch at the cursor frequency
Shift+F	DEMON carrier band 200-800 / 400-1400 / 1000-2000 Hz
Ctrl+F	Heterodyne shift 400/700/1000/1200 Hz
X / Shift+X (BB)	Broadband/fusion: TAS side of the selected contact flip / Shift: confirm
Z / X (TMA)	TMA page: hypothesis course -/+ 5 deg (Shift 1 deg)
Ctrl+Z / Ctrl+X (TMA)	TMA page: hypothesis speed -/+ 1 kn

Key	Action
Q / Shift+Q (TMA)	TMA page: hypothesis range +/- 1 NM (Ctrl 0.2 NM)
K / Shift+K (TMA)	TMA page: accept hypothesis as fix / Shift: copy solver proposal (training)
Shift+T (TMA)	TMA method: hypothesis/residuals, Ekelund range, dot stack (Shift+K in Ekelund: copy its range)
SPACE	LOFAR peak hold on/off
T	TMA for selected contact on/off
C	Classify contact (submarine / warship / biological / vessel / aircraft / torpedo)
G	Release / withdraw selected contact to CIC independently of classification
M	Set selected contact as target

## Standard procedure

1. Go slow (4-8 kn) and select HMS; stream the TAS at 3-12 kn (Y) and lower the VDS below the layer at 3-15 kn (Shift+Y).
2. Search BROADBAND for bearing traces; confirm tonals on LOFAR (N notch).
3. Select the contact (Up/Down), lock focus (Enter), check DEMON.
4. Classify (C); after a course change fit a TMA hypothesis (page TMA: Z/X, Ctrl+Z/X, Q) and accept it (K).
5. Need depth or exact range: ping (Shift+A), then release (G) and mark target (M).

Combat situation:

1. A tube-flooding transient warns that a shot may follow: turn the bow sonar and towed array onto its bearing and get the countermeasures ready. A launch transient, high-frequency sonar pulses, or a new loud broadband contact without tonals and with fast bearing drift may be a torpedo. Classify it as Torpedo (C) and report the bearing to the Bridge immediately.
2. Keep focus on the hostile submarine so the torpedo wire datum stays fresh.
3. Ping only when you need depth for the shot or the contact is about to be lost: the submarine hears a ping out to 60 NM and starts evading.

## Pro tips

- Gain (I/O) changes display and audio only, not detection. Black level (Ctrl+I/Ctrl+O) and contrast (Shift+I/Shift+O) help faint traces stand out; Shift+C changes the phosphor colour.
- D or A/B/H choose broadband, filtered or heterodyne audition. Heterodyne shifts the low band up to about 700 Hz so low tonals become audible.
- The listening audio runs about one and a half seconds behind the display (Remote Crew browsers about two seconds) so a busy moment never interrupts it. After steering the listening bearing the old beam fades into the new one after that delay; the stream is not cut.
- Put the TAS below the measured layer to hear deep targets; keep the HMS for shallow ones. Both arrays run in parallel.
- The TMA page shows the closing rate derived from the accepted solution: positive means the target is closing.
- If the TAS and HMS disagree by 9 degrees or more, treat the contact as a possible ghost (the display flags it) and turn to resolve it.
- The towed array is a line: it cannot tell a bearing from its mirror about the cable. A contact heard only on the TAS is marked "TAS left/right ambiguous" with its mirror bearing and does not feed TMA; the display shows the side you choose (starboard by default). Turn 20 degrees and watch both traces: the true one stays continuous, the ghost jumps (the status then reads "turn made - compare traces"). On

the broadband or fusion page X shows the other side and Shift+X confirms the shown side; nothing is resolved for you. A wrongly confirmed side stays mirrored (the TMA residuals will show it; X reopens the choice). A hull-sonar bearing resolves the side by measurement. Bearings toward the cable ends (endfire) are also less accurate than broadside.

- Shift+F selects the DEMON carrier band (200-800, 400-1400 or 1000-2000 Hz): search the band where the cavitation noise is strongest. Ctrl+F sets the heterodyne shift (400/700/1000/1200 Hz) for listening to low tonals.
- In the contact analyser (F8) with a contact selected, Enter assigns the browsed catalog profile to that contact and Shift+Enter clears it. The assignment is your annotation: it is shown in the contact list and saved, and it never changes the contact's classification or weapon interlocks.
- A contact is lost 120 s after its last detection. Keep tracking weak contacts, or reacquire with a ping.
- Wrecks return real echoes without Doppler. A submarine lying still beside a charted wreck hides in that echo from a CW ping (750 m range cell); an LFM ping resolves about 8 m and can separate the submarine from the wreck. Suspect every wreck the enemy could have reached.
- The bathythermograph (E) measures to the seabed, at most 1500 m. Only after a measurement does the weather & sonar analysis (0) show the layer, the shadow zone below it and a SOFAR channel.
- The sonar never names a torpedo or a submarine. It reports what it hears: a mechanical launch transient (heard out to 35 NM), high-frequency seeker pulses (about 6 NM) or the short transient of a submarine flooding a torpedo tube and opening its outer door (a warning that a shot may follow: out to 8 NM, but slow, quiet flooding only within 1.5 NM; both shrink with own noise, sea state and sonar damage) as a bearing, held on the Bridge alarm for 60 s, and breaking-up noises when a hull sinks. The OPZ symbol of a sonar contact follows your classification only; an unclassified contact stays unknown.

## **Not modelled**

- T switches the solver behind the training aid; the automatic solver never writes a fix by itself.
- No selectable split-window normalisation (TPSW); use gain, black level and contrast instead.
- The baffles blind only passive listening; an active ping of the hull sonar still sweeps all round.

## 3 Weapons

---

### Purpose

Weapons control turns a sonar contact into a firing solution. It launches the frigate's wire-guided torpedoes and its own ASROC, drops depth charges, manages the helicopter's stores (buoys, lightweight torpedoes), streams the Nixie towed decoy and releases the AA gun.

### Displays and instruments

Page 1 (target) shows the chart with the selected contact, the torpedo depth and the fire-control readiness line. Page 2 (stores) lists tubes, reload timers, torpedo stock, Nixie state, helicopter stores and the torpedo setup line: selected type with its remaining stock, search pattern, seeker enable point and salvo size.

The readiness line is checked top to bottom; the first failed check is shown:

```
BLOCKED: NO TARGET          select or take a contact (M)
BLOCKED: AFFILIATION ...    operator marked it FRIEND/NEUTRAL
BLOCKED: NO RANGE           ROE STD needs ping/TMA/buoy range
BLOCKED: NOT CLASSIFIED     classify as submarine or warship (Sonar C)
BLOCKED: NO TORPEDOES / NO TUBE READY / SALVO LIMIT
BLOCKED: WEAPONS ROOM DAMAGED
WEAPONS FREE                -> T or Ctrl+Enter
```

Torpedo run, seen from above:

```
frigate ==wire==> . . . . /\\//\\ ( datum )
                  mid-course   snake search   seeker on
                  to datum     +/-15 deg      inside 1.2 NM
                  (wire update  -> homes on nearest
                  every 0.5 s)   candidate
```

- Two torpedo types share the two tubes (60 s reload). Mk1: 45 kn, 12 NM, wire-guided. Mk2: 55 kn but only 8 NM. The scenario stock (default 6) is split 2:1 between Mk1 and Mk2; W selects the type, and if no tube holds it a tube unloads and reloads with it (60 s).
- Search pattern (X): the snake (+/-15 deg about the datum course, default), a circle of 0.4 NM about the enable point, or a helix that opens from 0.15 NM by 0.15 NM per turn to 1 NM. The pattern runs only once the seeker is enabled and has not acquired.
- Seeker enable point ( , / . ): 0.6 to 3.0 NM from the datum in 0.2 NM steps (default 1.2 NM). Earlier enable finds a target that has moved off the datum; later enable keeps the weapon quiet longer.
- Salvo (Y): one torpedo, or two in a +/-8 deg spread with their own datums turned about the ship; a spread needs two loaded tubes of the selected type and counts against the doctrine limit.
- Preset depth 10-300 m (default 60 m). A wrong depth is a miss: take depth from a ping, not from TMA.
- The wire updates the datum from the contact's observed position. Without updates it becomes STALE after 3 s and BROKEN after 12 s; the torpedo then continues to the last datum.
- The seeker homes on the nearest candidate: that can be a decoy, a whale or a merchant ship. A civilian hit ends the mission.
- Salvo doctrine SHOOT-LOOK-SHOOT: at most 2 own torpedoes running.
- ASROC (A): 4 rounds per mission. The rocket flies at 500 kn to the target's observed position (1 to 10 NM, current range needed) and drops the helicopter's lightweight torpedo there, set to the preset depth. It needs the same target checks as the torpedo and counts against the doctrine limit.

- Depth charges (Z): 20 per mission, dropped as a pattern of 5 (three along the wake 20, 80 and 140 m astern, two thrown 70 m abeam), then 45 s to reload the rack. The ship must make at least 10 kn. The charges sink at 3.5 m/s to the preset depth (15-300 m) or the seabed; each 90 kg charge is lethal within about 25 m and still damages out to about 100 m. Submarines within 5 NM hear the detonation and evade.
- ASW rocket launcher (R, RBU/Bofors type): 36 rockets per mission, fired in salvos of 6, then 60 s to reload. An attack salvo goes to the designated target's observed position 0.4 to 3 NM away (current range needed, same target checks as the torpedo): one round on the aim point, five on a ring of 80 m around it. The rockets fly at 400 kn (about 9 s per NM), and each round sinks at 11 m/s to the preset depth (10-300 m) or the seabed; the 23 kg charges are lethal only within about 14 m and harm out to 60 m, so a rough or stale fix wastes the salvo. **Defence salvo** (Shift+R): six rounds in a line 0.3 to 0.8 NM out along the bearing of a torpedo warning at most 5 s old, set to 15 m; a round that goes off within 35 m of a running torpedo destroys it, and the log reports that the torpedo noise ended. Every submarine within 3 NM hears the rockets splash into the water: an AI boat evades at once, the crewed submarine's sonar room reports the splashes with their bearing.

Both pages are laid out as a fire-control desk: on page 1 each tube has a lamp (green loaded, amber reloading, dark when empty) and the interlock chain (target, fix, ROE, weapon, flak) is a column of lamps that lights green stage by stage. Between the solution and the tube lamps an engagement sketch, north up around the own ship, draws the reach of the selected torpedo type as a dashed amber ring, the bearing to the target (dashed while it is bearing only) and, once a range is known, the estimated target position, the intercept point from the TMA course and speed and the torpedo run to it (green inside the reach, red beyond); the figure at the lower left is the radius of the sketch. It uses only the contact's observation, never the submarine itself; page 2 shows the remaining torpedoes, helicopter torpedoes, sonobuoys and RBU rockets as tank columns. The Remote Crew browser shows station, ROE and interlock as lamps and each tube as a column.

## Keys

Key	Action
M	Set target (from sonar contacts)
Hold Up / Down	Torpedo depth (10-300 m)
<- / ->	Select sonar contact for targeting
T / Ctrl+Enter	Launch torpedo (ROE check)
W	Torpedo type (tubes swap over; W cycles Mk1/Mk2)
X	Terminal search pattern: snake, circle, helix
, / .	Seeker enable point -/+ (0.6 to 3.0 NM, 0.2 NM steps)
Y	Salvo: one torpedo or two in a +/-8 deg spread
H	Launch / recall HSP-5
B	Deploy sonobuoys (HSP-5 airborne)
D	Lightweight torpedo from HSP-5
V	Stream one finite towed acoustic decoy
A	ASROC at the designated contact (range 1-10 NM)
Z	Depth-charge pattern over the stern
R	ASW rocket salvo at the target (fresh range fix, 0.4-3 NM)
Shift+R	Rocket defence salvo along the torpedo warning
Q / E	Zoom chart in steps, 500 to 0.5 NM
K	Camera follow on/off
F	Toggle AA gun fire release (withheld = never fires on raiders)

## Standard procedure

1. Take target from sonar (M) or select a contact with Left/Right.
2. Check the readiness line: range fresh, classification, tubes, salvo limit.
3. Set torpedo depth to the pinged target depth (hold Up/Down).
4. Fire (T or Ctrl+Enter); keep the contact fresh so the wire datum updates.
5. Defence: stream the Nixie early (V); launch HSP-5 (H) for buoys and air torpedoes.

### Combat situation:

1. Enemy torpedo reported: stream the Nixie at once (V). It lasts 600 s on a 0.2 NM cable; one ready, a second after 60 s. It runs at 10 m at 15 kn, deeper and closer astern when you slow down, and its cable parts above 25 kn. In a turn the cable lags behind.
2. Keep the counter-attack going: a fresh contact keeps the wire datum on the submarine.
3. With the helicopter airborne, a lightweight torpedo (D) can reach a distant contact faster than the ship's torpedo.

## Rules of engagement

ROE	Requirement
STD (start)	Current range (ping, TMA or buoy) and classification submarine or warship
FREE	Classification only; without range the torpedo is aimed 10 NM down the bearing

HQ switches to FREE by radio after the first hostile submarine is sunk; the player cannot change ROE. A contact marked FRIEND or NEUTRAL in Operations, directly or through a fusion, can never be engaged.

## Pro tips

- Fire from inside about 6-8 NM: at 45 kn the torpedo needs 8 minutes for 6 NM, and the submarine hears the launch out to 35 NM and starts evading.
- Aim the datum ahead of a moving target by keeping TMA running; the wire follows the observation, not the truth.
- Keep own speed below cavitation while guiding; losing the contact means losing the wire datum.
- The wire is a physical cable: it snaps if the ship runs faster than 20 kn or turns faster than 1.5 deg/s for about 5 s, when the ship-side spool (5 NM of own track) runs out, or when the torpedo has run 1.25 x its range.
- Torpedo range comes from its battery: at full speed it runs the catalogue range, hard manoeuvring throttles it and saves energy; when the battery is empty it coasts for a few seconds and is lost. Turning is slower right after launch (constant turning circle), and depth changes need a moment to build up.
- The warhead has a proximity fuze: it fires at the closest approach inside its radius, and the damage falls with distance (shock factor). A near miss can leave a submarine damaged but able to escape.
- A running torpedo is heard by the target through the sonar equation: quiet submarines in calm water hear it from a few miles, rain and their own speed mask it.
- Only two Nixies per mission: stream the first when a torpedo is likely, keep the second for the next attack.
- Homing seekers lock on the loudest candidate and only switch when another is clearly (6 dB) louder. They ignore echoes without Doppler, so a hovering target is hard to find; a torpedo that overruns a decoy without a hull hit remembers it and re-attacks. Hostile submarine decoys fade as their battery drains; hostile warships stream their own decoys when they hear your torpedo launch.

## Not modelled

- Depth charges only from the stern rack and throwers; the rocket launcher has no contact fuze (every round goes off at its set depth) and no anti-torpedo projectile of its own.
- One torpedo type for the helicopter; the doctrine limit of two own torpedoes running is fixed.
- No depth ceiling difference between Mk1 and Mk2; both run at the set depth.

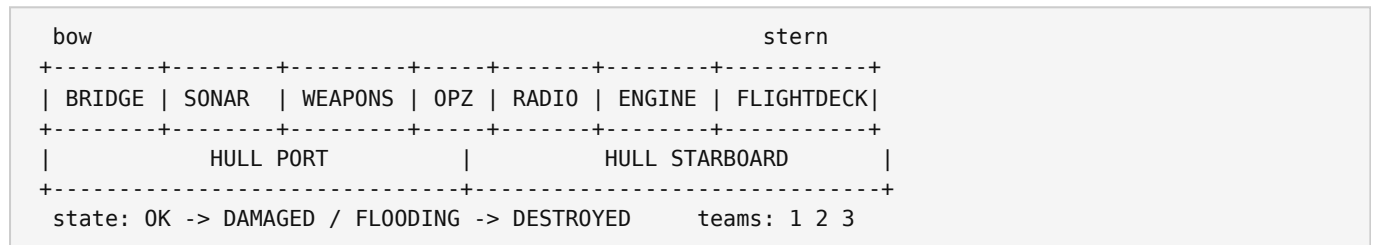
## 4 Damage control

### Purpose

Damage control keeps the ship afloat and the stations working after a hit. Three repair teams fight flooding and fire in nine compartments. Every compartment houses a station; a damaged compartment degrades it, a destroyed one disables it for the rest of the mission.

### Displays and instruments

Page 1 is the ship schematic, drawn as a damage-control console: water rises in each compartment from the keel, a fire glows red and a destroyed compartment is hatched. Each compartment's card carries a state LED, the flood and fire values with their LEDs and bars, and numbered badges for the teams on scene; a legend under the plan explains the LEDs. Page 2 lists details per compartment (flooding, fire, trend, teams on scene, heel); page 3 is the crew's watch bill. In the browser the Damage card opens with an annunciator panel (fires, flooding, lost, getting worse, total damage, list, trim, counter-flooding, teams busy, ship sunk) above a side view of the ship, bow to the left, and gauges for list, trim and total damage; a click on a compartment sends the selected team there.



- **FLOODING:** a hole below the waterline lets water in. The inflow follows the water pressure: fast at first, slower as the water inside rises towards the outside waterline. Rooms high in the ship (bridge) do not flood through a hole. At 70 % the compartment is DESTROYED.
- **DAMAGED:** the hole is patched; a small residual leak remains until a team pumps the room dry.
- **Patch kits:** plugging a hole uses one of 8 patch kits. Without kits a team can only pump against the open hole.
- **Hit location:** the torpedo's impact point decides the compartment; a close burst tears a bigger hole than a distant one. Missiles hit above the waterline and mostly start fires.
- **Fire:** grows with the room's fuel load (engine, flight deck and magazine burn fiercest) and is smothered by rising water. A room that stays hot for about 30 s ignites its neighbours. A flooded switchboard (sonar, operations, radio, engine) shorts and starts an electrical fire. A fire above 90 % in the weapons room cooks off the magazine: the room is destroyed and the neighbouring rooms are holed.
- **Stability:** floodwater adds weight, and loose water surfaces reduce the metacentric height (GM). The ship sinks when the floodwater exceeds its reserve buoyancy, and capsizes when GM is lost or the heel passes 35 degrees.
- **Heel:** off-centre floodwater lists the ship to that side and pulls it off course.
- **Counter-flooding:** with C and a list of 5 degrees or more, damage control opens the flooding valve of the high hull side; water enters at 0.5 % of the room per second until the list is cancelled, never past 60 % of that side, and the valve closes by itself below 1 degree (or with C again). The water is real floodwater: it adds weight and draught and a team has to pump it out later.
- **Trim:** floodwater forward or aft trims the ship (bow down counts positive). Every degree costs 0.5 kn of top speed and, bow down, adds own noise at the bow sonar. The stability line on page 2 shows list, trim and the open valve.
- **Steering gear and stabilizers:** the steering gear sits aft under the flight deck. If that compartment is destroyed, the rudder jams at its last angle until the room is repaired. A destroyed hull

compartment on either side knocks out the fin stabilizers, so the ship rolls more in a seaway. Floodwater adds weight: the ship sits deeper and accelerates more slowly.

Station effects: a station loses capability continuously with flooding and fire in its room (sonar and radar range shrink gradually); a destroyed room disables it. A damaged engine room caps speed at 15 kn, a destroyed one at 8 kn; a damaged or destroyed weapons room blocks torpedo launches; a destroyed flight deck prevents helicopter launch and recovery; a destroyed operations room also disables ESM.

## Crew and watches

Page 3 (Crew) shows the watch bill. The ship's company stands in three watches: one is on duty and tires, the other two rest and recover. The duty watch is relieved automatically every hour of game time (a real watch is four hours; the game compresses it so a session sees the rotation), or earlier with W on this page. For the first minute after a relief the new watch settles in and works at 85 %.

- **Fatigue** (0 to 100 % per watch) rises by about 25 % in a normal hour on duty and falls again at rest. Up to 25 % it costs nothing, so normal rotation keeps the crew at full performance. Above it, every 10 % costs 6 % performance.
- **Action stations** (G here or on the Bridge) put every watch on duty: the crew is 10 % more alert, but nobody rests and everybody tires in 90 minutes from fresh to exhausted, faster while fighting fire or flooding. There is no relief at action stations; standing down hands the watch to the freshest section. Stand to for an attack and stand down afterwards: after about half an hour at action stations the bonus is used up.
- **Morale** starts at 70 %. A sunk submarine (+15), a rescue (+12) or another task done (+6) raise it; a compartment damaged (-6), a task failed (-8) or declined (-2) lower it; a repaired compartment gives +2. Low morale tires the crew faster; each 10 % of morale changes performance by 2 %.
- **Performance** is what the crew delivers: the sonar operator needs a stronger signal (every 10 % lost raises the recognition threshold by 1 dB, so the passive ranges shrink), the lookout needs more contrast to sight, recognize and identify, and the repair teams patch, pump and fight fire at that pace. Page 3 lists the current values; the status ticker shows CREW with the performance, or GQ at action stations.
- **Wounded:** hits, fire, flooding and gas wound people. A torpedo hit wounds three at once in the station of the compartment it struck and one in the next; a missile hit or another sudden jump in damage wounds one for every 15 % of it; every minute in a fully burning or flooded compartment wounds one more (proportionally less for smaller fires). The sonar room (sonar and bridge compartments, 4 posts), the weapons (weapons and CIC, 4 posts) and damage control (all other compartments, 8 posts) each lose a post per wounded; a station with every post empty works at half pace, so an empty post in the sonar room raises the recognition threshold, empty weapons posts slow the tube reload and empty damage-control posts slow the repair teams (on top of the watch's performance). Every third wounded man is seriously hurt and out for the mission. Page 3 lists the wounded, the empty posts per station, the medical team and the spare hands.
- **Medical team** (M): it treats the lightly wounded of one station and sends one back to his post every 90 s. Without an order it goes where most are lightly wounded; M sends it to the next station with lightly wounded, after the last back to automatic.
- **Re-manning** (U): up to two men from the resting watches fill empty posts of the worst-hit station at once (serious cases first). They lose their rest: every resting watch gains 8 % fatigue per man. The ship has 6 spare hands for the whole mission, and the next re-manning takes 3 minutes to organise.
- The crewed opposing submarine has its own watch bill with the same rules and its own wounded (see the reference chapter); AI submarines re-man a station with two empty posts on their own, and the AI hunter frigate does the same.

## Keys

Key	Action
<- / ->	Select compartment
Up / Down	Select team 1-3 (unassigned)
Enter	Assign selected team to selected compartment
Backspace	Withdraw selected team
C	Counter-flood the high hull side against a list (again: close the valve)
W	Relieve the duty watch now (Crew page)
G	Action stations on/off
M	Medical team to the next station with wounded (Crew page)
U	Men from the resting watches to the worst-hit station (Crew page)
1-9	Always switch station; never assign a team
Click	Select compartment or its label; Enter assigns the selected team

On the uConsole the joystick buttons 1-3 assign team 1-3 directly to the selected compartment.

## Standard procedure

1. After a hit, read the compartment plan: fire, flooding, destroyed.
2. Fire first: select the room (Left/Right) and a free team (Up/Down).
3. Assign with Enter; watch progress and heel on the detail page.
4. Then flooding in machinery and sonar spaces, then the hull.
5. Withdraw teams (Backspace) once a room is stable and reassign.

## Pro tips

- Teams start in the operations room and need about 20 s per compartment to walk to their job; a team is only effective once it has arrived. Keep a team near the engine and weapons rooms.
- A team first patches the hole, then pumps. Patch kits are limited: spend them on rooms below the waterline, not on rooms that already stopped flooding.
- Two teams on one room halve the time. Teams cannot be assigned to a destroyed compartment.
- Fire in a room next to the engine or the weapons room is the most dangerous: it spreads into mission-critical spaces.

## Not modelled

- No named crew members or sleep by the clock; the wounded are counted per station (sonar, weapons, damage control), the bridge, engine room and flight deck have no posts of their own, and a wounded man never dies. Empty posts do not slow an AI submarine's sonar (only its torpedo gang and repairs). Watches are relieved every hour of game time.
- Counter-flooding only between the two hull sides; no selective flooding of other rooms.

## 5 Operations / CIC

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### Purpose

Operations (OPZ / CIC) builds the tactical picture above the water: surface and air radar, AIS, released sonar and ESM bearings, manual fusion of reports, NATO affiliation and air defence. It hands designated tracks to Sonar and Weapons.

### Displays and instruments

Page 1 is a full-height free chart with all published tracks; page 2 is the target page for the selected track; page 3 commands the patrol aircraft. The ship-centred radar picture uses its own range scale (10/20/40/80/120 NM, PgUp/PgDn), independent of the chart zoom (wheel, down to 0.25 NM radius; drag pans; K follows). Own units come from the datalink, not from sensors: the ship, the airborne helicopter ("HSP-5 DL") and every own weapon under way, i.e. torpedoes from ship, helicopter or ASROC (T<n>), ASROC in flight and ESSM, each with a friendly symbol and a heading tick.

```
NATO frame colours (operator annotation, not truth)
  yellow = UNKNOWN   blue = FRIEND   green = NEUTRAL   red = HOSTILE

report sources:  radar  AIS  sonar(released)  ESM  HFDF  lookout
track list:  ID  source  bearing  range  course/speed  age  class
```

- **Surface radar:** 30 NM, limited by the radar horizon (20 m mast) and target height; submerged submarines are invisible.
- **Mast and snorkel echoes:** a submarine at periscope depth with a raised mast or snorkel head (a crew's mast, a snorkelling submarine, one at radio depth, or an AI reconnaissance submarine's periscope during its look round) returns a tiny echo: in calm sea about half the sweeps find it at 7 NM, at sea state 3 at about 2.5 NM, at sea state 5 under 1 NM. It shows only as a bare dot that glows for about 6 s, with no symbol, no label and no track. Click the dot on the PPI or press B (newest dot) to mark it: a radar track R-... starts from that measurement and further echoes of the same mast update it; without new echoes it fades after 30 s. Blips and the mark are not saved.
- **Air radar:** 100 NM for aircraft and anti-ship missiles (ASM).
- The antenna turns once every 4 s: a contact is updated only when the beam sweeps past it, and each sweep detects it with a probability that falls with range (50 % at the nominal range for a broadside ship; bow-on targets are seen later, fluctuating echoes can miss a sweep). Sea clutter grows with sea state (about -5 % at sea state 4, -25 % at 6) and rain attenuates the echo (-10 % surface, -20 % air); from sea state 5 measurement errors increase. Inside 3 NM the CIWS search/track radar holds an inbound missile continuously while CIWS is released.
- **AIS:** civilian ships broadcast course and speed every 2-10 s (3 min at anchor) and their name about every 6 min. The VHF receiver hears them only within line of sight (about 20 NM). A radar track of a civilian shows name and course only after the matching AIS report has been received; radar alone gives position only. Optional live AIS/ADS-B traffic is indistinguishable from simulated traffic.
- **Fusion:** mark 2-8 raw reports (Space) and fuse them (L) into one operator track; Shift+L dissolves it. A fusion whose reports come from exactly one sonar contact can be designated to Weapons; its classification counts for fire control unless Sonar has classified the contact itself, and its affiliation applies to that contact. A fusion lasts only while all its reports are current.
- **Automatic fusion:** once a second the OPZ fuses reports of different sensors that lie on top of each other by itself, so one ship seen by radar, lookout and AIS is one contact. It uses the same gates as the correlation suggestions below, but only for a clear match: the score must be well inside the gates, at least one of the two reports needs a position (two bearing-only reports, such as sonar and ESM, are never fused automatically), and neither may have a second candidate from the same kind of sensor (two ships close together stay apart and appear as a suggestion). A further report joins an existing

automatic fusion the same way, up to 8 reports. An automatic fusion keeps its ID while at least two of its reports are current and drops a report that has lapsed; with fewer than two it ends. It takes its name from an AIS report, if it has one. Shift+L separates it, and those reports are not fused again automatically until one of them is new. With the OPZ down nothing is fused.

- **Sources:** reports inside a fusion are no longer listed or drawn on their own; the fusion stands for them (H shows them with the suppressed reports). Each row of the page 1 track list ends with sensor tags: R radar, V lookout, A AIS, E ESM, S sonar, H helicopter, B buoy, M patrol aircraft, F HF/DF, J home-on-jam, D datalink. Page 2 shows the full names under **Sources** for a fusion (for example Radar · Lookout · AIS). The Remote Crew OPZ station lists the sources of each fusion and hides its reports unless **Manage suppressed reports** is on.
- **Correlation suggestions:** the OPZ compares its current reports (at most 30 s old) from different sensors: sonar (only the ship's own bearings and fixes, not buoys or the dipping sonar), radar, ESM, lookout and AIS. Received AIS reports are OPZ reports of their own: the ship's reported position dead-reckoned to now by its reported course and speed, and its name once the static message is in; they count while their data is fresh (up to 10 min at anchor). Two reports whose bearings from the ship agree within 1.5° plus both reports' bearing uncertainties (together at most 8°) and, if both have positions, lie within 1.5 NM plus a tenth of their range of each other are suggested as a pair, provided their motion and class agree too: when both give a course and one moves at 3 kn or more, the courses must agree within 35°; when both give a speed, within 4 kn plus a quarter of the faster; two operator classifications must be equal, and an AIS report never pairs with a report classified as submarine, biological or aircraft. Agreeing classes rank a pair higher. Sonar and AIS are never paired with an air track. The sidebar of page 1 lists the two best (for example > K03 + R-2 Brg 087°); U fuses the top one exactly like marking both and pressing L, Shift+U dismisses it. The Remote Crew OPZ station lists all of them, with the course and speed difference and whether the classes agree, and the buttons **Fuse suggestion** and **Dismiss**. At most 4 suggestions exist at a time, each report in only one; reports already fused are left out. Suggestions are hints, not identification, and like fusions and dismissals they are not saved.
- **Suppression:** Delete hides a report locally; H shows suppressed reports again.

## Patrol aircraft

Page 3 commands a maritime patrol aircraft (MPA) on call from the nearest friendly airfield (without one it comes in from the nearest map edge). It flies at 300 kn in transit and orbits its search area at 200 kn in a 3 NM circle. Each sortie lasts up to 5 h including a 15 min reserve; at bingo fuel it turns home by itself. After landing it needs 30 min on the ground and then flies once more: 2 sorties per mission, each with 16 sonobuoys and 2 lightweight torpedoes.

- A requests the aircraft (it first heads for the ship's position) or sends it home.
- W sets the search area on the selected track's plotted position (without a selection on the ship); a click on the chart sets it on that point. A bearing-only track has no position to fly to.
- Z plans a buoy pattern (field, barrier, circle) about the search area; the aircraft flies the points and drops a buoy at each. Shift+Z cancels the pattern. X drops one buoy where the aircraft is, Y switches its buoys between PASSIVE and ACTIVE.
- T switches the aircraft's surface-search radar. From 300 m it sees ships and surfaced or mast-raised submarines out to 60 NM (limited by the radar horizon); its contacts appear as RADAR-MPA tracks with the aircraft as observer. AI submarines with a raised mast hear it and go deep (see the helicopter chapter).
- V starts or ends **MAD passes** (browser: **Start MAD passes*/End MAD passes**) while the aircraft is on its way or on station: once there it descends to 60 m and flies straight passes at 180 kn through the search area, turning back 2 NM past it (a cloverleaf). A submerged hull within about 400 m slant range is detected on a stateless draw each second (sure inside 250 m) and reaches the ship over the datalink as a MAD position fix without depth or course on that submarine's sonar contact. A buoy pattern flies first; A (home) ends the passes.
- D drops a torpedo on the designated sonar contact. The same checks as for the helicopter apply (current contact classified as a submarine, rules of engagement, a fresh fix under standard ROE), and

the aircraft must be within 2 NM of the datum.

Everything the aircraft learns reaches the ship only by datalink, out to 250 NM. Its buoys report only while the aircraft is within 50 NM of them; once it leaves or lands they go silent for the ship. The sidebar shows its state, bearing and range, the time left on station, stores, sorties left and how many of its buoys are being relayed.

## Keys

Key	Action
Up / Down	Select CIC track
C	Classify OPZ radar/HOJ reports or manual fusions
F	Set NATO affiliation
Shift+F	Cycle OPZ contact-domain filter
J	Enter a shared operator-visible track ID
Space / L / Shift+L	Mark raw reports and create/dissolve a fusion (Shift+L); matches fuse automatically
U / Shift+U	Fuse the top correlation suggestion (Shift+U dismisses it)
Delete / H	Suppress/restore locally; H manages suppressed reports
M	Pass CIC track to sonar/weapons
Page Up / Down	Radar range 10/20/40/80/120 NM
<- / ->	Select ASM track
E / Ctrl+Enter	Launch ESSM (VLS cell)
G	Deploy chaff (8 NM cone, cooldown)
R	Surface-search radar on/off (EMCON)
Shift+R	Air-search radar on/off (EMCON)
I	Toggle CIWS fire release (withheld = never fires on inbound ASMs)
Backspace	Clear all marked reports
B	Mark the newest bare radar blip as a track (or click the blip)
Enter	Confirm attack after classifying a real contact hostile
K	Camera follow on/off
A	OPZ page 3: request the patrol aircraft / send it home
W	Search area on the selected track (else own ship); click the chart for a point
Z / Shift+Z	Cycle buoy pattern about the search area / Shift cancels
X	Drop one buoy at the aircraft
Y	Aircraft buoy mode PASSIVE / ACTIVE
T	Aircraft surface-search radar on/off
V	Aircraft MAD passes over its waypoint on/off (low and slower, radar horizon shrinks)
D	Torpedo on the designated contact (aircraft within 2 NM of the datum)

## Standard procedure

1. Set radar range (PgUp/PgDn); switch radars for EMCON (R / Shift+R).
2. Select tracks (Up/Down), classify (C) and set NATO affiliation (F).
3. Reports on top of each other fuse by themselves; mark others with Space and fuse with L, Shift+L separates.
4. Designate the track for sonar/weapons (M).
5. Missile threat: chaff (G), CIWS release (I), then ESSM (E).

Air defence sequence (missile inbound):

40 NM	ASM detected (air radar / ESM seeker bearing)	
30 NM	ESSM envelope	-> E / Ctrl+Enter (2 fire channels)
8 NM	chaff cone	-> G (40 % break-lock, short blindness)
1.5 NM	CIWS	-> must be released with I

1. Air radar on (Shift+R), select the ASM track (Left/Right).
2. Chaff and manoeuvre first, then ESSM. Only 6 ESSM are loaded.
3. Keep CIWS released while missiles are inbound; withheld CIWS never fires.

## Pro tips

- Radar is a transmission that hostile ESM can intercept. Switch radars off (EMCON) when stealth matters more than the air picture.
- The radar does not know what an air contact is. Threat evaluation flags an air track as a possible missile (ASM) only from its own measurements: faster than 300 kn at or below 150 m, or a jamming strobe; a low, fast attack aircraft can raise the same flag. The flag needs about a second of plots, CIWS and ESSM engage only flagged tracks, and HFDF fixes and unclassified sonar contacts carry no domain until you classify them.
- Anti-ship missiles skim at about 20 m (5 m in the last 5 NM): radar sees them only inside about 20 NM, and a jamming missile gives only a home-on-jam (HOJ) bearing until it burns through. Missiles fly inertially to their launch datum, then their seeker needs the ship inside its cone for 1.5 s before homing; attack aircraft pop up to about 300 m for a few seconds to lock their fire-control radar before each salvo (an ESM warning and an early radar contact).
- Chaff lays a cloud beside the ship that blooms in about 3 s and drifts with the wind; fire it early enough for the cloud to bloom. CIWS must first slew onto the missile and kills mostly in the last few hundred metres.
- Affiliation is your annotation. Marking a contact, or a fusion containing it, FRIEND or NEUTRAL blocks every torpedo shot on it.
- Chart symbols follow NATO style on the uConsole and on every Remote Crew map: the frame shows your affiliation (hostile diamond, neutral square, friend wide rectangle, unknown without frame), the inner glyph the observed domain.
- J gives a track a shared ID that the whole crew (and Remote Crew browsers) sees.
- Enter confirms an engagement against a live (real-world traffic) contact after you classified it hostile; nothing fires automatically on unclassified contacts.

## Not modelled

- The helicopter's buoys belong to the helicopter station; OPZ handles only the patrol aircraft's buoys.
- The patrol aircraft has no dipping sonar and no own ESM; it cannot be shot down. MAD passes fly over the search area only, not along a track.
- Automatic fusion only for a clear match of reports from different sensors with at least one position; bearing-only pairs and ambiguous matches wait for the operator. Signatures are compared only as

the operator's classifications (no acoustic or emitter fingerprint matching), and AIS carries no ship type.

- No link-based air control of friendly aircraft.

## 6 Radio

### Purpose

The radio room handles communications with HQ and HF direction finding (HFDF). HQ sends orders, weather bulletins and ROE changes by teletype. HFDF takes bearings on submarines that transmit on HF or run a snorkel mast, out to 120 NM, far beyond sonar range.

### Displays and instruments

Page 1 lists current HFDF signals and the bearing log; page 2 is the teletype with HQ traffic; page 3 lists HQ tasks.

```
HFDF SIGNALS          BEARING LOG
> HF-03  247.0  age 12 s  HF-03 247.0 from pos A  t=12:04
  HF-05  061.5  age 40 s  HF-03 239.5 from pos B  t=12:08
                           -> CROSS-FIX + error ellipse on chart

pos A *-----_
      --_ X  <- fix
pos B *-----/
```

- Bearing error is +/-8 degrees for a ground wave and +/-16 degrees for a sky wave; signals older than 30 s can no longer be logged.
- Each signal shows its frequency and propagation. A submarine calling a distant shore station picks a high frequency by day (ground wave heard to about 95 NM) and a lower one at night (about 150 NM). Beyond the skip distance, several hundred NM away, the sky wave arrives instead. An AI submarine at periscope depth that has held the ship within the last 10 minutes reports it to its headquarters once every 30 minutes, at an unpredictable moment, with a 20 s call that HF/DF hears like any other.
- Logged lines and cross-fixes appear on the charts of Bridge, Weapons and Helicopter.
- In the Remote Crew browser the radio room has no chart: an HF/DF bearing scope (one strobe per signal, fan width is the bearing error, logged bearings dashed), receiver channels with frequency, propagation, signal meter and a log button, and the teletype. Selecting a channel opens the contact detail for annotation; logged bearings and cross-fixes are listed in the station panel.
- A second bearing of the same signal gives a cross-fix if it is taken at least 1 NM away from the first and within 300 s.
- The teletype also carries the weather bulletin every 30 minutes and HQ messages (threat warnings, ROE FREE).
- At mission start HQ reports the threat. With **coarse** intelligence it gives only a rough bearing and range of one threat. With **exact** intelligence it also names every hostile unit type committed to the mission with its number (for example "1x Altmetail (Diesel, älter), 2x air raid wave with anti-ship missiles"), using the names in the unit analyser (F8); positions stay unconfirmed. Patrol always gets exact intelligence, Double hunt and Nuclear intercept coarse, and the free hunt lets you choose on its difficulty screen (last row, "HQ intelligence").

Page 1 also shows an HF/DF bearing rose: each current signal is a strobe, fanned as wide as its bearing error.

### HQ tasks

Besides the hunt, HQ radios tasks to the ship: the first about 15 to 25 minutes into a built-in mission, then one every 25 to 45 minutes, at most six per mission and two open at a time. Custom missions get

none. Each offer arrives on the teletype and on page 3 (Tasks). Answer it within 5 minutes with A (accept) or D (decline); no answer counts as declined. A destroyed radio room cannot answer.

- **Distress call (SAR):** a life raft with 2 to 6 people, reported by EPIRB with about 0.5 NM error and drifting with current and wind. The survivors last according to the sea temperature, from 40 minutes in water below 8 °C to 100 minutes above 20 °C. The raft is sighted within 2 NM by day (3 NM at night by its strobe); then the circle on the chart shrinks onto it. Take them aboard by lying within 0.25 NM at 3 kn or less for 4 minutes, or let the helicopter hover overhead (one minute per person, only when the weather allows dipping). +600 points, -400 if they are lost.
- **Identify merchant:** HQ names a merchant within 60 NM and gives its position with about 2 NM error. It counts as identified once the lookout has published its identification or the helicopter passes within 1 NM with at least 1 NM visibility. About a third are flagged as suspect: HQ then passes a submarine datum near the ship. 40 minutes.
- **Submarine datum:** a circle of 5 NM radius from a maritime patrol report; not every datum has a submarine behind it. Search 10 minutes inside the circle with the ship or the helicopter. 50 minutes.
- **Replenishment at sea:** offered when fuel is below 70 % or torpedoes, ASROC or depth charges have been used; R on the Tasks page (or the browser's *Request supply ship*) asks for one yourself when at least 5 % fuel or any store is missing, at most once every 20 minutes after the last one ended. A friendly supply ship appears 18 to 28 NM away at 12 kn; its course and a dead-reckoning line are plotted. Keep within 0.3 NM and within 3 kn of its speed: fuel flows the whole time (a full load in 15 minutes), and torpedoes, ASROC, depth charges, Nixie decoys and CIWS and gun rounds come over in five loads, one every 3 minutes, each a share of what is still missing. Breaking away keeps what came over. The Tasks page shows what is aboard. VLS cells are not reloaded at sea. Worth +100, no penalty.
- **Radar silence (EMCON):** both radars off within 90 s and silent for 20 to 30 minutes. +200, -250 if a radar radiates.

Accepted positions are plotted on every chart (also in the Remote Crew browser). Scores are listed at mission end. The radio operator in the browser answers with the same buttons.

## Incidents at sea

The sea brings surprises of its own: the first 20 to 40 minutes into a built-in mission, then one every 30 to 50 minutes, at most four per mission (none in custom missions and lessons). Each is reported on the teletype.

- **Drift net:** a fishing boat reports a net 2 NM long across the ship's track, 3 to 7 NM ahead, hanging from the surface down to 20 m; the radio operator plots it on every chart as a ruler NET n, and it is hauled in after an hour. Running over it tears it: the fishermen claim damages (-100 points), and a streamed towed array or variable-depth sonar fouls in it and is hauled in at once. A submarine that crosses it shallower than 20 m fouls it too and is loud for 20 s while it tears free; deeper it passes under.
- **Weather front:** HQ warns 10 minutes ahead; then rain, a storm or fog holds for 30 to 60 minutes (visibility, wind and rain noise for every sensor, on both sides), and HQ reports when it has passed.
- **Merchant without AIS:** a freighter running without AIS appears 8 to 15 NM away; HQ reports it with about 2 NM error and offers it as an identify task when fewer than two tasks are open.
- **Whales:** a fishing boat reports a pod of two to four whales 3 to 6 NM ahead; they are real biological contacts for every sonar.

HQ passes the net, the front and the whales on to the submarine's broadcast; a crewed submarine hears of them when it copies the next broadcast, and its crew plots the net on its own chart.

## Own calls to HQ

On the Tasks page the radio room can call HQ itself, at most once every 10 minutes; the line at the foot of the task list says whether a call is on the air, how long until the next one, or that both are ready.

- **K Contact report:** sends the position of the freshest located contact (ping, TMA, buoy or fused fix, else an HF/DF cross-fix up to 15 minutes old). HQ acknowledges it on the teletype and vectors the patrol aircraft to it when the aircraft is airborne. HQ never says whether a submarine was really there: each report that put a hostile submarine within 3 NM of the fix earns 150 points at the mission's end (at most three).
- **H Request support:** HQ sends the on-call patrol aircraft toward the ship when it is available (even with the OPZ down), otherwise it says that no support is available.
- Each call is 20 s of HF transmission. While it is on the air, a submarine with its antenna up (the crewed boat's raised mast, an AI boat at periscope depth) takes an HF/DF bearing on the frigate (+/-8 degrees for a ground wave, +/-16 degrees for a sky wave): the crewed boat gets a report and a bearing line on its chart, an AI boat remembers the direction. Talking to HQ costs silence.

## Keys

Key	Action
Up / Down	Select HFDF signal
Enter	Log bearing with own position
Up / Down	Select HQ task (Tasks page)
A	Accept the selected task
D	Decline the selected task
R	Request a supply ship from HQ
K	Contact report to HQ (the freshest fix; HF call, can be DF'd)
H	Request support from HQ (the patrol aircraft; HF call, can be DF'd)

## Standard procedure

1. Read the teletype on page 2: HQ orders, weather, ROE changes.
2. On HF activity select the signal (Up/Down).
3. Log the bearing (Enter); the line appears on the chart.
4. Move at least 1 NM and log a second bearing within 300 s.
5. Hand the cross-fix to Sonar/Bridge as a search datum.

## Pro tips

- Take the two bearings from positions across the expected bearing line: the more they cross at right angles, the smaller the error ellipse.
- A submarine that transmits or snorkels is usually shallow and slow: a good moment to close in with the helicopter.
- Combine an HFDF bearing with a sonar bearing for a quick position estimate.

## Not modelled

- No free-text radio transmissions: the only own calls to HQ are the contact report and the support request, besides answering tasks; no communication plan or crypto.
- No frequency tuning: HFDF monitors the whole HF band and lists the detected signals with their frequency.

## 7 Engine room

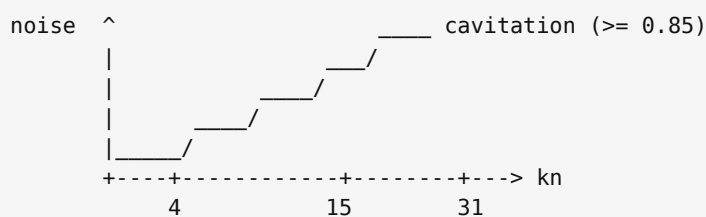
### Purpose

The engine room sets the propulsion order and manages the ship's acoustic signature. Speed is the most important trade-off in ASW: fast to reach a datum, slow and quiet to hear anything.

### Displays and instruments

The engine room is a machinery control console. Page 1 shows the engine telegraph as a column of lit steps, a large speed gauge (ordered speed as an amber mark, the damage speed limit in red), gauges for shaft RPM and own noise (cavitation zone in red) and lamps for shaft, plant, course, acoustic mode, cavitation and speed limit. Page 2 **Systems** has an annunciator panel of status lamps (dark when off, green while running, amber for a caution, red for an alarm) with a master lamp counting alarms and cautions, the fuel bunker as a tank column with stock, burn, endurance and range, gauges for roll, pitch and hull list, and a mimic of the ship's sections from bow to stern between the starboard and port hull, each with its water level, state, flooding and fire LEDs and the numbered repair teams at work.

TELEGRAPH	kn	own noise	
FLANK	31	#####	cavitating
FULL	16	#####	cavitating above 15 kn
> HALF	10	####	
SLOW	6	##	
STOP	0		
(ASTERN	3 kn, separate state)		



- Own noise rises linearly from 4 kn to 31 kn. The propellers cavitate when the blade-tip speed is too high for the water pressure at the screws: in calm water from 15 kn, in heavy seas earlier when pitching lifts the stern. Cavitation raises noise to at least 0.85 and cuts passive sonar range to 35 %.
- The lamp of the ordered speed is lit (ASTERN in amber). When a direct speed (V on the bridge) lies between two steps, a warning line names the ordered speed, so HALF 10 kn with 12 kn ordered is never mistaken for HALF.
- QUIET mode reduces own noise to 65 % and limits speed to 12 kn.
- Plant selection (G): AUTO runs the plant as before. DIESEL is the quiet plant (own noise about -4 dB, fuel -10 %) but caps speed at 18 kn; TURBINE gives full speed at about +3 dB and +25 % fuel. The choice is shown on page 2 and in the browser's engine room.
- Shaft RPM follows the fixed-pitch propeller: about 5.8 rpm per knot at steady speed (146 rpm at 25 kn, 181 rpm at the 31 kn flank speed). While accelerating the control programme keeps the shaft at most about 11 rpm ahead of the present speed; when slowing down the pitch reverses and the shaft idles at 20 rpm. The own shaft line on LOFAR moves with speed.
- Machinery damage caps speed at 15 kn (damaged) or 8 kn (destroyed).
- Fuel burn follows the power the propellers deliver: at steady speed it grows with the cube of speed, accelerating and braking cost extra. A lighter ship (burnt fuel) accelerates slightly faster; floodwater makes it slower and deeper. Heavy seas add resistance and cost up to about 1 kn at FULL. With empty tanks the shaft stops and no engine order is accepted.

- **Engine-room console (browser):** in the browser the Engine room's picture area is a machinery control console. An **annunciator panel** of status lamps (dark when off, turquoise while running, amber for a caution, flashing red for an alarm, each with its value) shows shafts, plant auto, diesel, gas turbine, silent running, cavitation, fuel, speed limit, machinery state, flooding and fire in the engine room, repair teams, fires and flooded sections aboard, grounding, towed array, sea state and roll; the master lamp in its plate counts the alarms and cautions. Below it are round **gauges** for speed (with the ordered speed as an amber mark and the damage speed limit in red), shaft RPM, own noise (cavitation zone in red), fuel, roll and pitch, the **fuel bunker** with stock, capacity, consumption, endurance and range, and the **ship sections** from bow to stern between the starboard and port hull strips, each with its water level, state, flooding and fire lamps and the repair teams at work. The console only shows; the orders stay in the station panel on the right.

### Keys

Key	Action
+ / -	Engine order (telegraph)
Up / Down	Telegraph up / down
A	Quiet mode QUIET/NORMAL
G	Propulsion plant: AUTO, DIESEL (18 kn, -4 dB) or TURBINE (+3 dB, +25 % fuel)
U	Enter target course directly (000-359)
V	Enter target speed directly (0-31 kn)

### Standard procedure

1. Choose the order with Up/Down or +/-; V enters an exact speed.
2. Stay below 15 kn while listening: cavitation masks the sonar.
3. Switch QUIET (A) for the attack approach (max 12 kn).
4. Never exceed 20 kn with the TAS streamed.
5. Watch machinery damage: speed limits of 15 or 8 kn apply.

### Pro tips

- Speed changes take minutes; order slow-down early before a listening leg.
- The towed array can only be streamed or recovered between 3 and 12 kn; at more than 20 kn with cable out it is lost.
- A hostile submarine hears you better than you hear it when you cavitate. Sprint only when you are far from the expected contact.

### Not modelled

- No individual shaft control; the plant choice applies to both shafts.

## 8 Helicopter deck

### Purpose

The HSP-5 "Sea Lynx" extends the frigate's reach: it flies to a datum at 120 kn, drops sonobuoys, dips its own sonar and attacks with lightweight torpedoes, while the frigate stays quiet and out of torpedo range.

### Displays and instruments

The station has four pages (8 again cycles them); it opens on page 3.

Page	Content
1 Status	Fuel, stores, weather limits, deck state
2 Mission	Chart with waypoint, buoys, contacts
3 Sonar	Dipping sonar and buoy contacts, depth, source
4 Acoustic	Listening: BROADBAND / LOFAR / DEMON of the dipping sonar or a passive buoy

frigate	waypoint (1-30 NM)
*----- 120 kn ----->	H hover + dip
buoy line (B): o ---- o ---- o	cable 15-300 m
PASSIVE: bearings;	))) dipping sonar
two crossing bearings ( $\geq 10$ deg) = fix	passive 18 NM
ACTIVE: range + bearing every 30 s	active 14 NM

- **Fuel:** 2 hours in forward flight; hovering (dipping) burns 1.3 times as fast. The helicopter returns automatically when only the 20-minute reserve remains. Running dry before landing loses the aircraft. In the hover the wind pushes it slightly downwind of its hover point.
- **Launch limits:** wind up to 32 kn, crosswind up to 22 kn, visibility at least 2 NM, sea state 5 or less, working flight deck, gusts up to 40 kn, cloud ceiling at least 300 ft, no severe icing, and a deck-motion window (roll within 8 degrees, pitch within 3.5 degrees). Light icing costs 20 % more fuel; the weather & sonar analysis (0) shows CLEAR, LIMITED or NO-GO. Landing also waits for such a window; turning into the sea reduces pitching.
- **Dipping sonar:** depth 15-300 m (default 75 m, at least 10 m above the seabed), passive 18 NM with  $\pm 2$  degrees, active ping 14 NM with 30 s cooldown. Dipping needs wind up to 30 kn, visibility of 1 NM and no icing.
- **Sonobuoys:** 5 per sortie, 8 NM range, 60 min battery; they drift with the current and a little with the wind. PASSIVE buoys give bearings (like DIFAR); ACTIVE buoys give range and bearing every 30 s (like DICASS).
- **Buoy patterns:** with X a pattern is planned: a queue of drop points about the waypoint: a 2x2 field (1.5 NM spacing), a barrier across the bearing from the ship to the waypoint (3 NM spacing) or a circle of 1.5 NM radius, each with up to 4 buoys of the remaining stock. The helicopter flies the points one after the other and drops the ordinary single buoy (in the selected mode) at each; SINGLE clears the queue, returning home drops it.
- **MAD run:** with Shift+M and the dipping sonar stowed the helicopter descends to 30 m and slows to 90 kn. A submerged hull within about 400 m slant range is detected on a stateless draw per sensor tick (sure inside 250 m) and reported as a MAD position fix without depth or course; it feeds the weapons' range check and, once the helicopter releases its contact, Operations.
- **Surface-search radar:** searches whenever the helicopter is airborne with the dipping sonar stowed (status line on page 2). From 150 m it sees ships out to 40 NM, surfaced submarines and raised snorkels or periscopes inside its radar horizon (about 30 NM). A mast is small: in calm water it shows

at about 10 NM, in sea state 3 at 3-5 NM, and in sea state 5 the clutter hides it. Every contact goes to Operations as a RADAR-HEL0 track with the helicopter as observer, one look every 2 s. A crewed submarine's ESM hears the radar and can warn its crew. Shift+R (browser: *Switch radar off*/*on*) switches the radar off and on again; switched off it neither sees nor radiates, and it stays off (saved) until switched on. An AI submarine with its mast or snorkel raised hears an aircraft radar within 40 NM (inside the radar horizon to its mast) on four of five 5-s looks, goes 40 m below snorkel depth and puts off snorkeling for 15 minutes while its battery holds more than 5 %; so a radiating helicopter drives snorkelers down, a silent one may catch them at the surface.

- **Lightweight torpedo:** 2 per sortie, 55 kn, 12 NM, dropped from the helicopter's position towards the datum, no wire. The target must be classified as submarine.

Page 3 shows the dipping sonar like a console: lamps for dome (green in the water, amber while lowering or raising), ping ready and water entry clear, a gauge of the cable in the water column down to the seabed, and a scope with the dipping and buoy bearings as wedges as wide as their error. Page 4 draws its waterfalls in the same phosphor colours as the ship's sonar.

## Keys

Key	Action
H	Launch / recall HSP-5
Arrow keys	Adjust waypoint bearing and range
M	Set sonar contact as air-torpedo target
B	Deploy one sonobuoy at current position
Shift+B	Next sonobuoy mode PASSIVE / ACTIVE
X	Buoy pattern: single, 2x2 field, barrier across the waypoint bearing, circle (X again: next; single clears)
Shift+M	MAD run on/off: low and slow, dipping sonar stowed
Shift+R	Search radar on/off (off: a submarine's ESM does not hear it, but it finds no masts either)
T	Sensor source: dipping sonar / sonobuoys
F	Confirm / unconfirm selected helicopter contact
C	Classify contact (submarine / warship / biological / vessel / aircraft / torpedo)
G / Shift+G	Select dipping-sonar contact / release or withdraw it from CIC
Y	Lower / retrieve helicopter dipping sonar
U / V	Raise / lower dipping-sonar target depth
A	Transmit active ping from deployed dipping sonar
D / Ctrl+Enter	Drop lightweight torpedo
Q / E	Zoom chart in steps, 500 to 0.5 NM
K	Camera follow on/off
Acoustic: Page Up / Down	Acoustic page: Broadband / LOFAR / DEMON
Acoustic: ← / →	Helicopter listening bearing -/+ 5 degrees
Acoustic: R	Reset listening bearing to automatic
Acoustic: T	Listening source: dipping sonar / passive buoys
Acoustic: J   , / .	Receiver audio on/off   lower/raise volume
Acoustic: I / O	Lower / raise gain (3 dB)
Acoustic: N	Notch filter against own propulsion
Acoustic: Shift+D	Cycle broadband / filtered / heterodyne audition
Acoustic: Shift+F	Cycle audition frequency band

Keys marked "Acoustic" apply only on the acoustic page (page 4).

## Standard procedure

1. Check weather limits and fuel; launch HSP-5 (H).
2. Place the waypoint ahead of the suspected target (arrow keys).
3. Drop a passive buoy line (B); switch ACTIVE (Shift+B) for range.
4. Hover and dip (Y), set depth below the layer (U/V), listen or ping (A).

5. Classified submarine: mark target (M), drop torpedo (D); recall in time.

Attack sequence:

1. Localise with two passive buoys or an active buoy/dip ping until the contact has a fresh position.
2. Classify it as submarine (C) and set it as target (M).
3. Fly to the datum; drop the torpedo (D or Ctrl+Enter). Keep contact for a second drop if needed.

## **Pro tips**

- Put the dipping sonar below the layer to hear deep submarines. The dip gauge shows the layer at the helicopter only once the lowered dome has passed through it; before that it shows only the charted water depth.
- Lay buoys ahead of the target's estimated track, not on top of the last datum.
- F confirms a helicopter contact; Shift+G releases it to Operations like a sonar contact.
- On the acoustic page, T switches the listening source between the dip and each passive buoy.
- Recall the helicopter in time (H): landing needs a working flight deck, and stores are not replenished between sorties.

## **Not modelled**

- No frequency channel management for buoys.
- The helicopter radar has no power or sector settings, only on and off; an AI submarine hears it only with its mast or snorkel up.
- Only one helicopter.

## 9 Electronic warfare

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### Purpose

Electronic warfare (EloKa) listens passively for radar emitters (ESM) and, when ordered, jams them (ECM). ESM detects radars out to about 150 NM, far beyond own radar, without transmitting. It gives bearings and emitter parameters that point to a platform type - and warns when a missile seeker locks on.

### Displays and instruments

Page 1 lists intercepts; page 2 shows the evidence for the selected intercept (frequency, PRF, modulation, candidates, correlation).

```
INTERCEPTS                                status  threat  band
> E-07  brg 312  9.3 GHz  PRF 2.4k  NEW     HIGH   X
  E-04  brg 045  3.0 GHz  PRF 1.0k  TRACK   LOW    S

EVIDENCE E-07
frequency 9.3 GHz  PRF 2400 Hz  modulation pulse-doppler
candidates (ranked, not identified):
  1. missile seeker  2. fire-control radar
correlation: compatible with CIC track T-12 (time/bearing)
```

- Bearing accuracy is about +/-3 degrees. Intercepts are bearings, not positions.
- A rotating search radar reaches the ESM antenna with its main beam once per revolution; its side lobes are heard only close in. The evidence page shows the peak signal level, a range estimate that assumes the power class of the best candidate (a wrong candidate gives a wrong range) and the measured antenna scan period.
- By default (realism level Standard or Realistic, F10) the evidence page shows only the measured parameters and a library lookup: every emitter whose published frequency (and PRF) range contains the measurement, in name order, without score. Radar type, threat and range estimate are then yours to judge; C cycles the library in name order. At the Beginner level, candidates are ranked by frequency, PRF and modulation with a score, and radar type, threat and range estimate are filled in. A tie is not an identification.
- Correlation with radar or sonar tracks uses compatible time, bearing and observed position, never hidden identity.
- The emitter library includes the anti-ship missile seeker (9.0-9.5 GHz, PRF 1.8-3.2 kHz, pulse-Doppler). It radiates only in the last 18 NM and only once the sea-skimmer is above the radar horizon, and it matches an attack aircraft's fire-control radar just as well: the bearing trend and the air picture decide.
- ESM runs from the operations compartment: a destroyed operations room disables it.

Beside the intercept list a bearing rose shows every intercept as a strobe in its threat colour, and lamps show ESM, jammer, automatic ECM and audio.

## Keys

Key	Action
Up / Down	Select a visible passive ESM intercept
F / Shift+F / B	Cycle status / minimum-threat / frequency-band filters
C	Assign radar type and release current bearings to CIC; clearing the assignment withdraws release
J	Engage / release the directional ECM channel for the selected intercept
Shift+J	Cycle noise, RGPO, VGPO, or false-target ECM technique
A	Toggle automatic ECM prioritization and soft-kill coupling
M	Toggle local ELOKA contact sound

## Standard procedure

1. Watch new intercepts; filter by status, threat and band (F/Shift+F/B).
2. Select an intercept and compare candidate radar types.
3. Annotate the radar type (C); this releases the bearing to CIC.
4. Missile seeker locked: jam (J), choose the technique (Shift+J).
5. Use auto ECM (A) under saturation; jamming is a transmission.

## ECM techniques

Technique	Effect
Noise	Masks the victim radar with broadband noise
RGPO	Range-gate pull-off: drags the seeker's range gate away
VGPO	Velocity-gate pull-off: drags the Doppler gate away
False targets	Injects false returns

Automatic mode (A) picks targets and techniques and couples jamming with soft-kill (chaff) during a missile attack.

## Pro tips

- Keep ESM running even under EMCON: it is completely passive.
- An intercept that changes from search to a high PRF pulse-doppler seeker at a steady bearing is a missile about to attack: warn Operations immediately.
- Assigning a radar type (C) releases the bearing to CIC; clearing the assignment withdraws it again.
- Jamming is a transmission. Use it deliberately, not continuously.
- Your own radar is heard too: a submarine at periscope depth (mast up, about 18 m) intercepts it on its ESM and can pass the bearing to other hostile units over its datalink. Deep submarines neither hear radar nor receive the datalink.

## Not modelled

- No communications intelligence (COMINT) or radio intercept; HF signals are handled by the radio room (HFDF).
- No decoy launchers other than chaff and no towed radar decoys.

## Reference data

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All values are the defaults of the current game version. Custom difficulty and missions can change stocks and time limits.

### Units

Quantity	Unit
Distance, range	nautical miles (NM)
Speed	knots (kn)
Depth	metres (m)
Frequency	hertz (Hz)
Course, bearing	degrees true, 000 = north, clockwise
Time	simulation seconds; 1x only

### Sensors

Sensor	Range	Accuracy / note
Passive sonar (base)	20 NM	bearing only; HMS +/-6 deg, TAS +/-2 deg, VDS +/-4 deg
Active ping	18 NM (reference target, broadside-quarter aspect)	CW or LFM (W); range accuracy from pulse and SNR, depth +/-12 m; 30 s cooldown; heard to 60 NM
Dipping sonar	18 NM passive / 14 NM active	+/-2 deg
Sonobuoy	8 NM	60 min battery
Surface radar	30 NM (50 % per sweep)	4 s antenna revolution; radar horizon; no submerged contacts
Air radar	100 NM (50 % per sweep)	aircraft and missiles; jammers burn through close in
ESM	150 NM (main beam)	+/-3 deg bearing; level and range estimate
HFDF	120 NM ground wave at 15 MHz (about 95-150 NM by frequency)	+/-8 deg bearing (sky wave +/-16)
Lookout	12 NM surface, 5 NM surfaced sub, 20 NM air, 20 NM land	x0.25 (new moon) to x0.45 (full moon) at night; fog and sea state reduce; class at 2, type at 3.2 resolved cycles over relative size (tanker about 7/5 NM, frigate 5/4 NM, speedboat 3/2 NM by clear day)

## Weapons and countermeasures

System	Data
Frigate torpedo	45 kn, 12 NM (battery), wire-guided (ship $\leq$ 20 kn, $\leq$ 1.5 deg/s, 5 NM spool), 2 tubes, 60 s reload, depth 10-300 m, proximity fuze
Helicopter torpedo	55 kn, 12 NM, 2 per sortie, no wire
Hostile torpedo	40 kn, 20 NM, homes from 3 NM; faster than the frigate, so outrunning it alone rarely works; an AI submarine attacks a located frigate within 10 NM even when she runs quiet
Nixie towed decoy	2 per mission, 600 s, 0.2 NM cable (10 m at 15 kn, deeper when slower, parts above 25 kn), 60 s reload
ESSM	6 missiles, 30 NM, 2 fire channels
CIWS	1.5 NM, 180 rounds, needs release; 115 deg/s slew, own track radar inside 3 NM
AA gun	240 rounds, needs release
Chaff	6 rounds, 8 NM, 40 % break-lock when bloomed; cloud drifts with the wind for 90 s

## Own ship

Item	Value
Speed	4-31 kn; telegraph STOP 0, SLOW 6, HALF 10, FULL 16, FLANK 31 kn
Turn rate	about 0.075 deg/s per knot (1.2 deg/s at 16 kn); turning circle about 0.4 NM
Cavitation	from 15 kn in calm water, earlier in heavy seas; passive range $\times 0.35$
QUIET mode	noise $\times 0.65$ , max 12 kn
TAS handling	3-12 kn, stream 360 s, recover 480 s, fault above 20 kn
TAS depth	20-260 m, minus 4 m per knot
Damage	9 compartments, 3 teams (about 20 s walk per compartment), 8 patch kits; sinks beyond reserve buoyancy, capsizes at 35 degrees heel or lost GM

## Environment

Process	Model
Tide	M2 (12.42 h) + S2 (12 h), 0.4-1.4 m amplitude, larger in shallow water
Surface layer	seasonal base depth; about 8 m shallower in the afternoon; deepens in wind above 12 kn; internal waves +/- 6 m
Sound speed	Mackenzie equation from the temperature profile (sea surface 8-18 deg C by season)
Current	steady field up to 1 kn plus 3 % of the wind, 20 deg right of downwind
Seabed	rock, gravel, sand, silt or mud; affects bottom reflection
Hazards	up to 64 charted wrecks and submerged rocks (tops at least 15 m deep), shown on every chart (wreck: hull line with masts, rock: asterisk; depth of the top when zoomed in, details in the tooltip); both raise the seabed within their footprint and are obstacles for the ship, submarines and weapons
Shipping	cargo ships, tankers and passenger ships steam from destination to destination (the charted ports and exits at the edge of the sea area) on steady courses; they give way under the collision regulations (head-on, crossing from starboard or overtaking: alter 35 deg to starboard when the pass would be closer than 0.5 NM; any ship alters under 0.25 NM) and run from a detonation within 8 NM at full speed for 10 min; work boats and fishing vessels keep wandering, convoys and HQ task ships hold their course
Atmosphere	barometer 975-1025 hPa that falls ahead of rising seas; air temperature from the sea, season, day and cold northerly winds (below 0 deg C in winter storms: snow, icing); gusts; cloud ceiling; sun elevation with civil/nautical twilight; moon phase
Rain lens	rain freshens the top few metres (up to -1 PSU, mixed away by wind) and lowers the surface sound speed
SOFAR channel	an interior sound-speed minimum (about 400-500 m below the surface layer) exists only in deep enough water

## Weather & sonar analysis (key 0)

Key 0 opens a full-screen analysis panel over any station (0 or Esc closes it; the simulation keeps running). In the web client every station opens it with 0 or from the workstation menu. Over the ocean profile the mouse reads depth and sound speed; over the sound-path section it reads range, depth and sound speed and says whether that point lies in a shadow zone or a convergence zone.

- **Environment:** time, daylight (day, civil or nautical twilight, night), moon phase, weather and precipitation, visibility, wind with gusts and Beaufort force, sea state, barometer with its 3-hour tendency (rising, steady, falling, falling rapidly), air and sea temperature, cloud ceiling and icing. A rapidly falling glass below about 1004 hPa gives a storm warning. The weather system changes by at most one sea state per hour, so the barometer moves faster than a real one.
- **Weather effects:** sun (strong layer), wind (deeper mixed layer) and rain or snow (fresher surface water, rain noise) light up while they act.
- **Helicopter flight weather:** CLEAR, LIMITED (within 80 % of a limit, or light icing) or NO-GO, with wind, gusts, crosswind, visibility, ceiling, sea state, deck roll and pitch, icing and whether dipping is possible.
- **Submarine (crewed submarine stations, key 0 on the uConsole and the web client):** instead of the helicopter flight weather, cloud ceiling and icing, the submarine's stations show what the weather does to the submarine. *Mast on radar:* the range at which a surface-search radar like the frigate's detects a raised mast or snorkel head (half of all sweeps) in the current sea and rain, beside the calm-sea value; sea clutter hides the small echo, and the radar horizon caps it. *By eye:* how far

a ship's lookout sights the submarine surfaced in the current light, moon, visibility and sea (5 NM on a clear, calm day); a raised mast at periscope depth is never sighted by eye. *Ambient noise:* wind and rain noise above a calm sea (sea state 1) in the four sonar bands (100, 400, 1600, 6400 Hz); it masks the submarine from passive sonar and dampens its own listening alike. *Snorkel:* at most 6 kn, +12 dB radiated level and diesel lines at 50 and 100 Hz that a sonar can hear.

- **Ocean profile:** appears only after the sonar has taken a bathythermograph (Sonar E): measured sound speed over depth, the layer, a SOFAR axis if present, nine sound rays from the hull sonar to 20 NM and the shadow zone below the layer (red) where the hull sonar hears little. The measurement is marked stale after 30 min or 10 NM.

## Chart plot tools (key P)

The crew keeps one shared grease-pencil plot. Every station and every Remote Crew browser sees the same drawing, and it is saved with the game. It is the crew's own drawing: nothing in it comes from a sensor, and it never changes the simulation.

- **Opening it:** press P on the Bridge, Weapons or Helicopter map or on the OPZ chart. A cursor appears on own ship. Arrow keys move it (Shift: faster), or click on the chart. Enter sets a point, Esc cancels a started object and then ends plot mode, and P also ends it. A hint bar at the top of the chart shows the active tool and keys on the left and the cursor's bearing and distance from own ship on the right.
- **Tools:** M mark (one point); R ruler (two points, shows bearing and distance); B bearing line from own ship through the cursor (own position and time are stored, so the line stays where it was laid); C circle (centre, then a point on the radius, at most 200 NM); D dead-reckoning line (start point, then a point in the direction of travel, then type the speed 0-60 kn). The DR line moves on with time and shows its CPA to own ship's present course and speed.
- **Erasing:** Backspace deletes the object nearest the cursor. Shift+Backspace clears the whole plot.
- **Labels:** objects are numbered M1, R2, B3 and so on. In the web client you can type a label before drawing or rename an object in the list under the map.
- **Web client:** choose a tool above the map, then click once (mark, bearing line) or twice (ruler, circle, DR line). "Plot track bearing" lays the selected track's measured bearing from its observer position.
- **Limits:** at most 64 objects and 24 characters per label.

## Chart history and labels

Every tactical chart (Bridge, Weapons, Helicopter, OPZ, the submarine's chart and plot, and the Remote Crew charts) shows where things were, not only where they are:

- **Own track:** a faint dotted line behind the own ship or submarine, one point every 30 s of simulation time, the last 2 hours.
- **Contact history:** earlier positions of a track as small dots that fade with age, one every minute, the last 12 per track.
- **Bearing history:** for a bearing-only contact the chart keeps its last 6 bearings; the selected contact's earlier bearings are drawn dashed from where each was taken, so their crossing shows where it may be.
- **Labels:** chart labels move aside instead of covering each other, the own ship or a symbol: first to the right, then down, up and to the left; bearing-line labels slide along their line.

The history is display only: it is built from what the sensors reported, is never saved and forgets a track 15 minutes after its last report.

## Opposing submarines

Class	Quietness	Max depth	Torpedoes
Diesel (older)	0.75	200 m	4
AIP (modern)	0.85	250 m	5
Nuclear attack	0.92	400 m	8

Submarines evade for 240 s after hearing a ping or a torpedo, may launch a decoy, lie in wait, snorkel (detectable by HFDF and ESM) and sometimes ping from 15 NM or less. Near the frigate a submarine may instead creep to a charted wreck within 8 NM and lie still on the bottom beside it for 15-30 minutes. A submarine with its mast or snorkel raised that hears an aircraft radar (helicopter or patrol aircraft) goes deep and holds off snorkeling for 15 minutes. In the frigate scenarios (1 to 4) a patrol submarine that has heard no ping or torpedo for 10 minutes, keeps more than 2 torpedoes and is more than 10 NM from the frigate torpedoes a merchant passing within 4 NM on about one in seven of its once-a-minute fire windows; each merchant lost costs 300 points.

Submarine physics: the hull accelerates toward an ordered speed (no instant sprints); hydroplanes need speed (below about 4 kn depth changes are slow); radiated noise rises about 12 dB per doubling of speed and jumps when the screw cavitates, and the cavitation speed rises with depth; a torpedo launch makes an 8 s transient; a badly flooded submarine blows ballast once and rises fast and loud; operating below test depth fatigues the hull, and 1.5 x test depth crushes it; a lurking submarine holds its position against the current. Submarines sense like you do: passive bearings from their own sonar, a range only after their own TMA legs (a few minutes), and a shot on that TMA only once its range error has converged (the difficulty field "enemy fire-control convergence": sigma over range at or below 0.25 in the patrol scenarios, 0.15 for the SSN; a solution older than 90 s or re-opened by your course change is not fired on), ESM only with the mast up, the datalink only at mast depth or snorkelling, and a torpedo alarm takes the crew a few seconds (2-15 s) before the submarine evades. Surface ships lose top speed in heavy seas (small ships more).

## Crewed opposing submarine (Remote Crew)

A second crew can play the enemy. The submarine has seven stations: Command, Sonar, Weapons, Engine room, Mast & ESM, Navigation and Radio room; a browser takes them like the frigate's (F9). As long as one of them is held, the living hostile submarine with the lowest number follows only that crew's orders; when the roles are released or the host revokes them, the AI takes the submarine back from where it is. A save keeps the crew's orders, modes, mast, wires, plot, alarm bearings, ESM picture and radio log; after a load the submarine runs its last orders and waits up to ten minutes for its crew to take the stations again (every submarine station is re-leased) before the AI takes it back. A browser holds roles of one side only (frigate or submarine), never both; the lobby first asks which unit it plays. In solo mode the one browser holds all nine frigate stations, or with **Play the submarine** in the host bar all seven submarine stations (and back with **Play the frigate**). The host's **New game** dialog picks the **Side** too (*Frigate F-217* or *Submarine*); as the submarine the frigate, its helicopter and the patrol aircraft are run by the **AI hunters**.

- **Stations:** *Command* orders course, speed and depth, lies on the bottom, pings and takes a BT, and sees the whole submarine. *Navigation* orders course and depth, keeps the submarine's plot and watches keel and obstacles. *Engine room* runs the telegraph, snorkel, silent running, the trim and the emergency blow and watches battery and noise. *Mast & ESM* raises the mast and watches ESM and alarms; Command may raise it too, to look through the periscope. *Weapons* fires, guides the wires and launches decoys. *Sonar* is the submarine's sonar room. The *Radio room* copies HQ's broadcast and sends situation reports; it may raise the mast for its antenna. Each order is accepted only from the station that owns it.
- **Depth steps and displays:** Command and Navigation order depth in one step: periscope depth (15 m, keeps the mast usable), snorkel depth (submarines with a snorkel), above or below the layer (15 m

above / 30 m below; only after the submarine's own BT measurement, since the crew knows the layer only from it) and deep (the safe depth over the charted bottom). On the uConsole these are U, Shift+U, J, Shift+J and H. The browser shows the water column (surface, periscope depth, measured layer, ordered, safe and crush depth, seabed, the submarine and its dive direction), large readouts for course, speed, depth and battery with coloured mode and alarm chips, and an ESM rose with each emitter's strobe and the ping and torpedo alarm bearings; the uConsole's Mast & ESM page has the same rose. Mast, snorkel, silent running and lying on the bottom have separate on and off buttons.

- **Orders and weapons:** the submarine follows course, speed and depth orders within its turn, depth and acceleration limits; telegraph steps (stop, 3, 6, 10, 15 kn, maximum) set the speed quickly. Fires a torpedo down a sonar contact's measured bearing, with its ping fix or TMA solution while current, or down a free bearing with an optional range; firing needs a flooded, loaded tube and the target inside the tube arc. **Tubes:** the crew takes over with the loaded tubes flooded. A fired tube stays empty until the torpedo gang loads it from the racks, if any torpedo is left there (M at the Weapons station, browser: Load; every submarine carries at least as many reloads again as it has tubes, and loading takes 2 min on a nuclear submarine, 3 min on a conventional one and 4 min on the older diesel classes, slower in stale air); a loaded tube is dry and must be flooded before it fires, which opens its outer door, takes 20 s and is audible for 4 s like a short transient that the frigate's sonar hears out to 8 NM (Shift+M, browser: Flood); slow flooding takes 60 s and is heard only within 1.5 NM (Ctrl+M, browser: Flood quietly). The Weapons station and the browser list each tube as empty, loading, dry, flooding or ready, and the log reports every tube loaded and flooded. The AI's submarines load and flood by themselves: quietly and early once they hold a fix on the frigate within 15 NM, loudly and just before the shot when they must fire on dry tubes. The crew sets the run depth (5-300 m, otherwise a shallow default) and fires one torpedo or two in a  $\pm 4^\circ$  spread, each with its own datum. Every crew torpedo runs on a wire: the crew can move its datum (bearing and distance from the submarine) and the wire turns it onto the new datum until its seeker acquires; faster than 10 kn or turning harder than  $1.5^\circ/\text{s}$  for 5 s breaks the wire, as does running out of either spool, and the crew can cut it. Launches a decoy and blows main ballast in an emergency (three times on a full air store).
- **Plant and submarine modes:** a crewed submarine never goes up, snorkels or calls home by itself. The battery drains with speed and hotel load; below 20 % the log warns, and an empty battery limits the speed to what the plant can serve (an AIP plant still takes over the load by itself). **Snorkel** runs the diesels at snorkel depth and charges the battery, at most 6 kn; diving deeper shuts the head valve. The running diesels are loud: +12 dB radiated level, a lower quiet factor and two firing lines at 50 and 100 Hz in the submarine's LOFAR signature (AI submarines too). **Silent running** limits the submarine to 5 kn and makes it as quiet as a lurking AI submarine. **Lie on bottom** stops the submarine 3 m above the seabed where the water is no deeper than test depth: silent and no drift; any speed or depth order lifts off. The submarine stops short of land or a seamount instead of turning away, and the log warns in shallow water.
- **Energy and stores:** the engine room's **Energy & stores** (browser card, uConsole Engine room page 2 **Stores**) shows the energy balance at the present speed (load, supply and net kW, the time until the battery is empty or full), the battery and AIP oxygen, the diesel bunkers and a table of how long the battery lasts dived at each telegraph step and how far that carries the submarine (on the uConsole tank columns for battery, AIP oxygen, diesel and absorber and a bar per telegraph step). **Diesel:** the bunkers hold 300 hours of the generators' full output and a patrol starts with 65 %; only the running diesels burn it (0.27 l per kWh), the log warns at 10 %, and with dry bunkers snorkelling no longer charges. The **charge rate** sets what snorkelling does: *full* (the whole generator output, +12 dB and both diesel lines), *half* (half the output, +9 dB, weaker lines) or *air only* (the fans without diesels, +4 dB, no lines). **Air:** dived, the crew uses oxygen and breathes out carbon dioxide (about 0.45 % per hour each); a CO₂ absorber set takes CO₂ out until it is spent (8 spare sets), an oxygen candle adds 1 % O₂ over 15 minutes (12 aboard, one at a time), and snorkelling flushes the submarine toward fresh air within minutes. From 3 % CO₂ or below 18 % O₂ the air is stale, from 5 % CO₂ or below 16 % O₂ it is foul; the log warns at each step. Stale air slows the crew (down to 30 % performance), and the torpedo gang reloads accordingly slower. AI submarines manage their air by themselves and come up to air the submarine when it turns foul. A nuclear submarine has none of these stores.

- **Tanks, trim and air:** the engine room's **Tanks & trim** (browser card with a cross-section of the submarine, uConsole Engine room page 3 **Tanks**) shows the main ballast, the regulating tank, the trim tanks, the high-pressure air and the trim the submarine is in. Every weight change moves the submarine off neutral: a torpedo leaving a bow tube makes it 1.5 t lighter and bow light, water in flooded compartments (see damage control) makes it heavier and trims it toward the flooded end; the automatic trim takes up what its tanks can of that weight and moment. With the **automatic trim** on, the engineer pumps the regulating tank ( $\pm 8$  t, 25 kg/s) and the trim tanks ( $\pm 3$  t fore and aft, 15 kg/s) back to neutral; by hand, each order moves the regulating tank 0.5 t or the trim water 0.25 t (and switches the automatic off). Running trim pumps are audible (+3 dB and a 120 Hz line). Whatever the tanks cannot take up sinks or lifts the submarine by 0.03 m/s per tonne, and a trim angle ( $1^\circ$  per tonne of moment, + bow down) drives it down or up with speed; the hydroplanes hold that only with way on, so a heavy submarine hovering at low speed sinks below its ordered depth; the log warns from 2 t and from  $3^\circ$ . The **high-pressure air** (200 bar) holds three emergency blows of 60 bar each; a blow empties the main ballast in 20 s and the submarine rises to 10 m and stays there, the ordered depth reset to 10 m. The next order below 12 m opens the vents: the main ballast floods in 40 s before the submarine can dive. Snorkelling on the diesels runs the compressor (0.05 bar/s); with the air only on the fans it does not. Without power neither the trim pumps nor the compressor run. The AI's submarines keep themselves trimmed and keep their one legacy blow.
- **Damage control:** the engine room's **Damage control** (browser card with a damage-control lamp panel, a side view of the pressure hull with water, fire glow, gas haze, leaks, shut bulkheads, a state lamp per compartment and the team badges, gauges for trim, floodwater and high-pressure air, then the table; uConsole Engine room page 4 **Damage**) divides the pressure hull into six compartments: bow room, control room, quarters, battery room, engine room and stern room. A hit on the crewed submarine holes the compartment it strikes (a leak of 1.5 % per % of hit damage, up to a full hole; from 50 % damage the neighbour too, with half the leak) and may start a fire there (chance = damage / 150); a hull failure below test depth holes it as well (see Below test depth). Water comes in at 40 kg/s through a full hole at 100 m, growing with the square root of depth; above half a compartment it spills into open neighbours (20 kg/s) and smothers a fire. A fire grows to full in 2 min and then spreads through open bulkheads; seawater in the battery room (from 2 t) gives off chlorine gas that drifts through open bulkheads and clears slowly once the battery is dry. Water in the battery room (from 5 t) or a fire there cuts the **power**: the motor stops (no way on, so the hydroplanes do not hold a heavy submarine), and the trim pumps, compressor and electric bilge pumps stand still. **Shutting the bulkheads** of a compartment keeps water, fire and gas in it and starves a fire there in 3 min. Two **damage-control teams** walk the submarine (8 s per compartment) and **seal a leak** (a full hole in 60 s), **pump out** (30 kg/s, a quarter by hand without power) or **fight a fire** (a full fire in 60 s); in gas they work at half rate, and in a compartment 90 % full they can only pump. A compartment half full of water, half on fire or half gassed takes its station out: the bow room the torpedo tubes, the control room the mast and periscope, the engine room the diesels, the stern room half the top speed. The floodwater is weight and moment for the trim (see above); a submarine that sinks below 1.5 x test depth is crushed. The log reports leaks, fires, sealed leaks, fires out, flooded compartments, chlorine and power.
- **Engine-room console (browser):** in the browser the Engine room's picture area is a machinery control console instead of a chart. An **annunciator panel** of status lamps shows every plant state at a glance: dark when off, turquoise while running, amber for a caution and flashing red for an alarm, each with its value (motor, silent running, cavitation, snorkel, generator, battery, charging, fuel, oxygen, carbon dioxide, absorber, oxygen candle, main ballast, blowing, vents, high-pressure air, compressor, trim pumps, automatic trim, trim angle, power, flooding, leak, fire, gas, over depth, emergency ascent, on the bottom); the master lamp in its plate counts the alarms and cautions. Below it are round **gauges** for speed, battery, energy balance, depth (test to crush depth in amber), high-pressure air and trim angle with the ordered value as an amber mark, **tank columns** for battery, fuel, AIP oxygen, absorber, high-pressure air, main ballast and the regulating and trim tanks (these from their middle: up heavy, down light), and the **compartment mimic** from bow to stern with the water level, the leak, fire and gas lamps, the bulkheads and the teams at work. A nuclear submarine shows no battery, diesel or air stores. The console only shows; the orders stay in the station panel on the right.

- **Below test depth:** the crew may order the crewed submarine below its test depth, down to crush depth (1.5 x test depth); the "deep" step and lying on the bottom stay at the safe depth. The uConsole's and the browser's depth columns mark the crush depth, and while the submarine is below test depth Command on the uConsole shows a red **BELOW TEST DEPTH** alarm and the browser a red alarm chip, both with the test and crush depths. From 90 % of test depth the hull fatigues as before (one failure in about 30 min at test depth); below test depth failures come far faster, growing with the square of the excess: about one every 6 min at 110 %, one a minute at 125 % and one every 20 s at 140 %. Each failure is one of three: **sheared bolts** of a fitting (a quarter leak in a random compartment, 6 % damage), a **failed seal** of a shaft or valve (half a leak in the engine or stern room, 12 % damage) or a **cracked pressure hull** (a full leak and a 60 % leak in the neighbour, 30 % damage). Just past test depth a crack does not happen; its chance grows by 15 % per 10 % of excess, up to 60 %, and a seal fails in 30 % of the rest. The log reports every failure with its compartment, and at crush depth the hull collapses and the submarine is lost. The AI's submarines keep to their test depth.
- **Atmosphere:** the submarine has its own sounds, on the uConsole when it plays the submarine and in the submarine's browsers with sound on (never on the frigate's). From 60 % of test depth the hull creaks, now and then at first and every 4 s at test depth and below; a hull failure cracks. Every detonation in the water within 30 NM (a torpedo or missile hit, a merchant torpedoed) is heard: within 2 NM as a heavy blast close aboard, farther off as a dull distant rumble, and the log reports it with the bearing the crew's ears give (a few degrees off). A hunter's ping (hull sonar, dipping sonar or active buoy) rings on the hull. With stereo sound both come from that bearing, left for port and right for starboard of the submarine's head. In **silent running** the submarine rigs for red: the uConsole's submarine screens and the browser's submarine command stations turn to dimmed red light until silent running ends.
- **Crew:** the submarine has its own watch bill with the frigate's rules (see Damage control, Crew and watches): three watches, fatigue, action stations and morale. The Engine room and Command order action stations (B on the uConsole, a button in the browser's Damage control card) and relieve the watch (M on the Damage page, or the button). Morale rises when a ship sinks and falls with every 10 % of hull damage. A tired crew hears later on sonar, sights later through the periscope and its damage-control teams seal and fight fire more slowly (the pumps are machinery and keep their rate). **Wounded** follow the frigate's rules too: hull damage (one wounded per 15 % in a hit) and a minute in a fully flooded, burning or gassed compartment wound people in the control room (sonar, 4 posts), the bow room (torpedo gang, 4 posts) or the rest of the boat (damage control, 8 posts); empty posts slow sonar recognition, tube loading and flooding, and the repair teams. The log reports the wounded; the Damage page shows them with the empty posts and the spare hands. Shift+M sends the medical team to the next station, Ctrl+M re-mans the worst-hit station from the resting watches (the browser's Damage control card has both buttons).
- **Situation picture:** an intercepted active ping or torpedo is logged with the bearing the submarine's own ears measured (a few degrees off) and shown with its age in the alarms. At periscope depth the **mast** can be raised; its ESM then reports the radars sweeping the submarine with bearing (log and ESM list), and the mast lowers by itself when the submarine goes deeper. With the mast up the **periscope** page (Command page 3, Mast & ESM page 2; P raises the mast at both) shows the eyepiece in the start screen's look: sky and sea in the light of the hour (day, dusk, night with stars and the moon), clouds, rain, snow and fog from the weather, the horizon moving with the sea, a true-bearing scale and a crosshair; the scope trains in 2° steps (←/→, Shift: 10°). Everything the optics make out within the frigate lookout's contrast model at 2.5 m eye height (day/night, moon, visibility, sea state, land in the way) appears as a silhouette and as a bearing-only **sighting** with a coarse class (warship, merchant ship, vessel, aircraft, torpedo wake) and its apparent length; the log reports each new sighting. Neutral ships show their navigation lights here as on the bridge's binoculars. Made-out ships, submarines and aircraft appear as turned 3D models as in the binoculars, at the full length their apparent length and the judged angle on the bow give, each as the model of its real type; the crew's sighting still names only the coarse class. ↑/↓ tilt the head 2° (Shift: 10°, from 10° down to 60° up, for aircraft), ,/. switch between low power (32° field) and high power (8°), and Space switches the stabilizer; tilt and field stand in the corner of the picture. These settings belong to the eyepiece of the uConsole or of each browser, change only the picture (not what the optics detect) and

are not saved. A helicopter in sight hangs at its true elevation above the horizon in the still sky, behind the clouds. Underway the water streams past just below the eyepiece with the submarine's own speed (toward the eye looking ahead, from bow to stern looking abeam). Enter reads the **stadimeter** on the sighting under the crosshair: the range follows from its apparent length and the assumed hull length of the class (130 m for a warship or an unrecognized vessel, 150 m for a merchant), so an unrecognized or bow-on target reads long; the reading is  $\pm 25\%$  and becomes a VISUAL fix on the submarine's sonar contact of that target for 120 s, usable for a shot like a ping fix. Aircraft and wakes cannot be ranged. Each reading is also a **mark** for the **attack computer**: from two or more marks at least a minute apart (the last six within 15 minutes) it fits the target's course and speed in a straight line and, with the torpedo's speed, shows the lead angle (left or right of the bearing) and the running time under the periscope (browser: the Solution column); the quality grows with the time between the first and last mark (full at 5 minutes) and the number of marks. Ctrl+Enter on the periscope page (browser: Fire on solution, Command only) fires on the solution of the sighting under the crosshair: the torpedo runs on the intercept course to the point where target and torpedo meet. A shot at the sonar contact of a marked target uses the solution as well. A target turning after the last mark leaves the solution behind; a faster fit than 40 kn is rejected as a bad mark. The marks are part of the saved game. The charted coast stands on the periscope's horizon as far as its low optics see land (hills assumed 25 to 70 m). Command can also ping and take a BT without a sonar operator. The chart shows the tube firing arc where the tubes cannot fire all round, and the sonar room's assigned target is preselected for the shot.

- Mast & ESM:** with the mast up at periscope depth the submarine's own ESM antenna (3 m above the water) hears the radars around it once a second: the frigate's, other ships' and aircraft radars inside the radar horizon, over land only where the coast does not block the line. The frigate's helicopter radiates its X-band search radar while it flies and is not dipping (antenna at 150 m), the patrol aircraft its frequency-agile search radar while the OPZ has it switched on. Each intercept carries the measured bearing ( $\pm 4^\circ$ ), band, carrier frequency, PRF, modulation and received level, never the emitter's identity or position. The crew keeps an **emitter list** (E1, E2 ...) across mast periods: an intercept joins an emitter when bearing, band and waveform agree (a frequency-agile radar by bearing and band only), and the list forgets an emitter 30 minutes after its last intercept. **Classification** is the crew's annotation from the library: the emitters whose published frequency and PRF ranges hold the measurement, up to 16, best fit first: each entry shows its fit (**good**, **fair** or **poor**: frequency and PRF near the middle of its ranges and the same modulation fit best), entries of one grade stand by catalogue key, and the list only reorders when a grade changes; the choice sets the power class for the **range estimate** from the level (unclassified: the shortest range the library allows). Every 30 s each emitter keeps a **bearing** from the submarine's own position (20 minutes); the chart shows the latest bearing lines, and once the submarine's own motion has swung the bearing by at least  $8^\circ$ , the **cross-fix** is their best crossing with a 95 % error ellipse that allows for an emitter drifting up to 8 kn since each line (a fast frigate usually gives none; a fix whose lines disagree is marked). The **level trend** reads rising, steady or falling over five minutes. The **scan period** is the time between an emitter's main-beam hits (the level peaks; close in, the weaker side lobes are heard in between and do not count), measured only on the emitter's own waveform, over gaps up to 12 s with no washed scan between them, and read after 8 s: **rotating** with its period (about 2.5 s for a navigation or surface-search radar, 5 s for an air-search radar, 2 s for a multi-function radar) is a search radar sweeping past; **steady** (the beam on the mast at every scan) is a tracking or fire-control radar. A steady, live emitter is always a mast warning, and the log reports once when an emitter turns steady. The **mast warning** ("radar can see the mast") comes when a live search radar's estimated range is inside the range at which a surface radar sees a raised mast in this sea and rain (the weather page's value); the **recommended mast time** is 60 s in a calm sea, up to 300 s when sea clutter hides the mast and 20 s under that warning, and the log reports when it is exceeded. From sea state 3 waves wash over the antenna and some scans hear nothing. The browser's Mast & ESM card has the rose, the emitter table and the selected emitter's evaluation; **Transfer to plot** puts the cross-fix (mark and error circle) or else the latest bearing line into the submarine's plot. On the uConsole's Mast & ESM page ↑/↓ select an emitter, ←/→ step through its library classification and Enter transfers it to the plot.
- Radio room:** fleet headquarters sends a submarine broadcast every 10 minutes (broadcast 0 at mission start, then 1, 2 ...) and repeats it until the next one. The submarine copies it only with its

antenna up, which is the raised mast at periscope depth, and only after 20 s of unbroken reception inside that broadcast's time on the air; a submarine that stays deep misses broadcasts and only ever gets the latest. A broadcast carries HQ's **contact report** on the frigate in 60 % of cases: a position 5 to 15 minutes old with an error circle of 4 NM, and its course and speed rounded. On the chart it is drawn as an amber circle with its age, and the radio page gives bearing and range from the submarine. A **situation report** (Enter at the radio room, or the browser's button) is 20 s of HF transmission with the antenna up. During it the frigate's HF direction finder can take a bearing on the submarine (radio room HF/DF), and lowering the mast aborts it. HQ acknowledges a report in its next broadcast and then always adds a sharper contact report (2 NM). The radio page and the browser's Radio room card show the antenna, the broadcast number and time to the next one, copy and transmit progress, the reports sent, the latest contact report and the radio log (12 entries, saved). Below the mast the **VLF loop antenna** still copies the broadcast down to 25 m, but the slow VLF signal needs 60 s of unbroken reception instead of 20 s; it only receives, a situation report still needs the mast. From broadcast 2 on, a broadcast may carry an **HQ order** (half of them, while no order is open, at most 4 per mission; a missed broadcast is a missed order): proceed to an area 8 to 15 NM away in deep water and reach it within 3 NM inside 40 minutes (a green circle on the chart), send a situation report within 30 minutes, or keep radio silence (no transmission) for 20 minutes. The radio page and the browser's Radio room card show the open order with bearing, range and time left, and how many were carried out; the log marks the order a broadcast brought. Orders do not decide the mission. A broadcast also passes on the incidents at sea HQ knows of (drift net, weather front, whales; see the Radio chapter); the crew plots a net on the submarine's chart. With the antenna up the radio room also hears the frigate calling HQ (a contact report or a request for support, 20 s each): it reports the HF/DF bearing (+/-8 degrees ground wave, +/-16 degrees sky wave) and draws a 30 NM bearing line labelled HF on the chart.

- **Counter-detection and evasion:** the submarine's sonar room tells an intercepted active ping by its frequency as the frigate's hull sonar, a helicopter's dipping sonar or a sonobuoy, and logs it with the measured bearing and the received level (dB re 1  $\mu$ Pa, from source level, spreading and absorption); a buoy pinging within 12 NM is heard only by a crewed submarine. A buoy splashing into the water within 4 NM is heard with a rough bearing ( $\pm 6^\circ$ ); a salvo from the frigate's ASW rocket launcher splashing in within 3 NM is reported with its bearing while its charges sink (about 11 m/s), time to change depth or get out from under them; and a torpedo alarm is kept with its bearing. The **Threat** page (Command page 4 and Navigation page 2; the browser's Threat card at Command and Navigation) counts the pings of the last 5 minutes by source, gives the loudest level with its assessment (from 150 dB the pinger probably holds an echo) and the trend of the last two pings (rising: closing in), splashes, torpedo alarms and ESM emitters, the submarine's own signature (depth against the measured layer, own noise: cavitating, snorkelling, loud, moderate or quiet, the mast) and the crew's recommendations. The chart draws each intercept of the last 2 minutes as a dashed bearing ray. **Evade** (I at Command or Navigation, the browser's **Evade** button) gives the crew's evasion order for the freshest alarm (under 2 minutes): against a torpedo, the smaller turn that puts it 150° on the quarter, full speed and a decoy; against active sonar, the stern to the pinger, silent running at 3 kn. Both cross the measured layer (down when above or in it, up when below it), or go deep without a BT; the submarine lifts off the bottom first. The order goes through the same checks as the single keys, and the page shows it before it is given. The picture is display only and not saved.
- **Navigation and plot:** the engine room has telegraph buttons, and Command and Navigation share the submarine's own grease-pencil plot (marks, rulers, bearing lines, circles, DR lines); the frigate never sees it, and the submarine's plot is not saved. The navigation display shows the water under the keel and checks the chart along the ordered course up to 5 NM: land or a seabed shallower than the submarine is reported as an obstacle ahead, in the log and as a warning. Only charted geography counts; other vessels are not in the check.
- **Submarine sonar:** the same sonar workstation as on the frigate (broadband, LOFAR, DEMON, TMA, active echoes, classification, listening audio), but the hull array listens at the submarine's own depth, so the layer works for and against the crew. There is no towed array and no release to an OPZ. An active ping gives echoes and is heard by the frigate.
- **What the submarine crew sees:** its own submarine, the known chart, its own sonar measurements and its own torpedoes in the water. It never sees the frigate's position, its plot, its events or its

mission messages; the frigate crew cannot tell a crewed submarine from the AI.

- **Mission:** unchanged. From the submarine's side it is won when the frigate sinks, when the submarine escapes (it leaves 150 NM from its start) or when it holds out to the time limit of a hunt; if the submarine sinks, its crew sees "Submarine lost". D on the submarine's end panel opens the submarine's own **debrief**: its track, what its sonar held on the frigate, the true positions of frigate, helicopter, patrol aircraft and buoys, torpedoes both ways, the pings it took and the spans (at least 1 min) in which the frigate's sonar really held it, marked below or above the layer. Spoken crew reports (Options page 2) speak the submarine's log when the uConsole plays the submarine: torpedo and pings heard, buoy splash, new contact, evasion, mast warning, leak, fire, action stations, each own torpedo away (the log names its tube), detonations close aboard or distant and breaking-up noises with their measured bearing, a copied HQ broadcast (with or without a contact report on the frigate), a new periscope sighting with its class and bearing, passing 90 % of test depth and going below it on the way down, a hit, every hull failure, and the mission won or lost; browsers at submarine stations get the same reports.
- **Saving:** the crew binding is not saved. After a load the AI commands the submarine until a crew takes its roles again, and the submarine's sonar picture starts empty.

## Playing the submarine on the uConsole

Every new game first asks **Which unit do you play?**: *Frigate F-217* or *Hostile submarine* (Up/Down or 1/2, Enter; the last choice is preselected). Outside a mission, Options (F10) page 2 **uConsole plays** changes it as well, for example before loading a game. With the submarine the uConsole commands the hostile submarine instead of the frigate. The frigate is then crewed from the browsers through Remote Crew (F9); every frigate station no browser holds is crewed by the **AI hunters** (below). The uConsole shows only the submarine's own picture; the frigate's banners, event log, sound cues, plot and tooltips never appear, and its trackball and telegraph controls are disabled. The side can only be changed outside a mission; it lasts for this launch and is never saved, so every launch starts with the frigate. A solo browser (Remote Crew solo mode) chooses the side in the same way: its **New game** dialog has a *Side* field, and with *Submarine* the session takes the submarine's seven stations while the AI hunters crew the frigate.

The top bar shows the submarine's seven stations as tabs: 1 Command, 2 Sonar, 3 Weapons, 4 Engine room, 5 Mast & ESM, 6 Navigation, 7 Radio room (Tab or a click on a tab switches), and on the right the mission, the clock, speed, course and depth. Each order key works only at the station that owns it, as in the browser; elsewhere a banner names the right station. Browsers can crew the submarine's other stations at the same time; a station a browser holds is marked in the top bar and is not operated from the uConsole. Every station but the sonar room is laid out like the Bridge: the chart on the left (known geography, the submarine's own plot, the ESM bearing lines and cross-fixes, the submarine with its ordered course and motion vector, the bearing lines of its own sonar contacts or their symbol while a ping or TMA fix is current, its own torpedoes T1..., a limited tube firing arc), the station on the right with a threat bar (torpedo alarm and active sonar heard with measured bearing, hull damage, cavitation, low battery, ESM radar intercept) and the station's page. The threat bar appears only while a threat is current: a heard ping or an ESM intercept fills it for 30 s; after that an amber triangle with the number of standing warnings sits in the top bar and the details stay on the Threat page. **Command** has four pages, the **Engine room** four, **Mast & ESM** two and **Navigation** two (the station's key again or PgUp/PgDn). The Engine room's second page **Stores** shows energy, endurance and air; R cycles the charge rate, A fits an absorber set and 0 lights an oxygen candle. Its third page **Tanks** shows the cross-section, main ballast and air, the tanks with their orders, weight, trim angle, drift without planes, flooding and pumps; ↑/↓ pump out or flood the regulating tank, ←/→ move trim water aft or forward and Z switches the automatic trim. Its fourth page **Damage** is a compartment mimic from stern to bow (water level, fire glow, gas haze, LEDs for leak, fire, gas and shut bulkheads, the water in tonnes, team badges), lamps with the selected compartment's water, leak, fire, gas, bulkheads and the power, and both teams; ↑/↓ pick a compartment, ←/→ a task, Enter sends team 1 and Shift+Enter team 2 there with that task, and I shuts or opens the compartment's bulkheads. **Navigation** (page and station) shows course and depth, water under the keel and any charted obstacle ahead, speed, own noise, battery and the active modes, below them a row of round dials for course (the ordered course as an amber mark), depth (amber beyond test

depth, red beyond crush depth) and speed as on the frigate's bridge, and the water column under the submarine: submarine depth, ordered depth, safe depth and seabed; the layer appears there only after the submarine's own BT measurement (E at the submarine sonar). **Weapons** (page and Weapons station) shows fire readiness, torpedoes, tubes ready (flooded), reload, decoys, emergency blow, a line with the tubes that are not ready (M loads the next empty tube, Shift+M floods the next dry one, Ctrl+M floods it slowly and quietly) and the submarine's own sonar contacts (bearing, range where known, and quality in percent: the better of the signal quality and the track confidence, 100 % being a firm contact), below them the engagement plot of the selected contact, drawn like the frigate's engagement sketch (torpedo reach, bearing, estimated position, intercept point and torpedo run, from the submarine's own observation only). The **Engine room** is a control console: round gauges for speed (the ordered speed as an amber mark), battery (depth on a nuclear submarine) and own noise, lamps for silent running, snorkel, on the bottom, cavitation, emergency blow and the plant state, and the telegraph steps; **Mast & ESM** shows the mast time, the rose, the emitter list and the selected emitter (signal, level and trend, cross-fix, classification); its second page and Command's third are the **Periscope** (eyepiece, line of sight, light and the sightings list; ←/→ train, Enter stadimeter). The **Radio room** has one page: antenna, broadcast schedule, copy and transmit progress, HQ's latest contact report and the radio log; Enter sends a situation report and P raises or lowers the mast. Below are the submarine log and the submarine's telemetry, as a band or a status ticker as set in the options; orders, shots and decoys are logged there.

Key	Action
1 ... 7 / Tab	Stations: 1 Command, 2 Sonar, 3 Weapons, 4 Engine room, 5 Mast & ESM, 6 Navigation, 7 Radio room; Tab next (click a tab too)
C / V / D	Order course / speed / depth (Command; course and depth also Navigation, speed also Engine room)
U / J / H	Depth steps: periscope / snorkel depth (Shift), below / above the measured layer (Shift), deep (Command, Navigation)
Page Up/Down	Command pages: Navigation / Weapons & contacts / Periscope / Threat (or 1 again); Navigation pages: Navigation / Threat (or 6 again); Mast & ESM pages: ESM / Periscope (or 5 again)
Q / E	Zoom chart in steps, 500 to 0.5 NM
K	Chart follows the submarine on/off
Wheel / drag	Zoom / pan the chart (mouse on the chart)
Arrow keys	Select an own sonar contact
Ctrl+Enter	Fire a torpedo at the selected contact (Weapons)
F	Fire down an entered bearing, then the distance to the datum (Enter: none) (Weapons)
X	Launch a decoy (Weapons)
M	Load the next empty torpedo tube (Weapons)
Shift+M	Flood the next loaded tube (20 s) and open its outer door; loud, the frigate may hear it; only a flooded tube fires (Weapons)
Ctrl+M	Flood the next loaded tube slowly (60 s); the frigate hears it only very close (Weapons)
Shift+B	Emergency blow, once (Command, Engine room)
T	Torpedo run depth 5-300 m (Weapons)
Y	One torpedo or a two-torpedo spread (Weapons)
W	Steer the newest wired torpedo: bearing, then distance (Weapons)
Shift+W	Cut the newest torpedo's wire (Weapons)
G	Silent running on/off, at most 5 kn (Command, Engine room)
Shift+G	Lie on the bottom / lift off (Command, Navigation)
N	Snorkel up/down, diesels charge at snorkel depth (Engine room)
P	Raise/lower mast at periscope depth: ESM hears radars, the periscope sees, the radio antenna is up (Command, Mast & ESM, Radio room)
Arrow keys	Mast & ESM page: select an emitter
← / →	Mast & ESM page: classify the emitter from the library (annotation, never truth)
Enter	Mast & ESM page: cross-fix (or the bearing line) into the submarine's plot
← / →	Periscope page: train the scope 2° (Shift: 10°) (Command, Mast & ESM)
↑/↓ · , / . · Space	Periscope page: ↑/↓ tilt the head 2° (Shift: 10°), , / . low/high power (32°, 8° field), Space stabilizer
Enter	Periscope page: stadimeter range of the sighting under the crosshair (Command, Mast & ESM)

Key	Action
Ctrl+Enter	Periscope page: fire on the attack computer's solution of the sighting under the crosshair (Command)
+ / -	Telegraph step faster / slower (Command, Engine room)
Sonar keys	As on the frigate sonar, without towed array, OPZ release, plot and telegraph
R	Engine room, stores page: cycle the snorkel charge rate (full, half, air only)
A	Engine room, stores page: fit a fresh CO2 absorber set
0	Engine room, stores page: light an oxygen candle
Arrow keys	Engine room, tanks page: flood (down) or pump out (up) the regulating tank
← / →	Engine room, tanks page: trim water forward (right) or aft (left)
Z	Engine room: automatic trim on/off
Arrow keys	Engine room, damage page: pick a compartment (up/down) and a task (left/right)
Enter	Engine room, damage page: send team 1 (Shift: team 2) with the task
I	Engine room, damage page: shut or open the compartment's bulkheads
I	Command, Navigation: evade the freshest ping or torpedo alarm (course, speed, layer, silent running or decoy)
Enter	Radio room: send a situation report to HQ (needs the mast up at periscope depth; the frigate can take an HF bearing)
M	Engine room, damage page: relieve the duty watch now
Shift+M	Engine room, damage page: medical team to the next station with wounded
Ctrl+M	Engine room, damage page: men from the resting watches to the worst-hit station
B	Action stations on/off (all watches on duty, alert but tiring)
Ctrl+B	Clear baffles: 60° to starboard for two minutes, then back (the hull sonar is deaf astern)
0	Weather panel of the submarine (0 or Esc closes)
S / L / F9	Save / load / Remote Crew (the frigate's crew)

## Submarine missions

Scenarios 5 to 7 give the submarine the objective; the frigate's task is to stop it. They are meant for a crewed submarine (the uConsole on the submarine side, a solo browser as the submarine, or a submarine crew in the browsers), but the frigate can play them against the AI too: an uncrewed mission submarine replaces its patrol legs with the objective's and still evades, lies in wait and counter-attacks as in any other scenario. It transits at 6 kn 30 m below the layer and detours round land and water shallower than 30 m. While it is hunted (a ping or torpedo heard in the last 10 minutes, or its own contact on the frigate within 12 NM) it creeps at 3 kn; it slips away from a ping at 5 kn below the layer (only a torpedo makes it run) and counter-attacks a located frigate four times as readily as a patrolling submarine. For the breakthrough it runs for the goal area and passes 40° wide of a frigate it holds within 6 NM of its leg. For reconnaissance it closes the frigate's last position it heard, else the position in HQ's latest contact report from the radio broadcasts, else the frigate's patrol area, leading a running frigate at 4 kn more than its speed; it comes to periscope depth at 3 kn within 8 NM and looks round with its periscope for 24 s every 90 s. The head sweeps from the bow clockwise in 16 s and makes the frigate out with the lookout's contrast model at 2.5 m eye height (light, moon, visibility, sea state, land in the way); the report counts as soon as a look has sighted the frigate within 8 NM. While the periscope is up it is a raised mast for the frigate's surface radar and the patrol aircraft (see OPZ, mast echoes). In the convoy

attack it lies in wait 2 NM ahead of the convoy and 3 NM abeam of its track, hovering at 2 kn; only a convoy that has passed it is chased on an intercept course at 4 kn more than the convoy's speed. Within 4 NM it turns its tubes on the nearest merchant and fires one torpedo at a time (at most one a minute); its torpedoes take merchants like the crewed submarine's. The submarine's orders stand in one line over its chart and as the mission line of the browser's submarine stations.

- **Breakthrough (5):** the goal area (3 NM radius) lies about 5 NM beyond the frigate's patrol position, seen from the submarine's start, in water at least 40 m deep; the submarine's chart marks it as GOAL and the orders give bearing and range from the submarine. The submarine wins when it enters the area, the frigate when it sinks the submarine or holds it off for 5 hours. The rule that a submarine escapes 150 NM from its start does not apply.
- **Reconnaissance (6):** the submarine must get the frigate in sight through the periscope and complete a situation report in the radio room while the frigate is still among its sightings. The orders read "Frigate in sight" as soon as it is. A report without the frigate in sight does not count. The frigate wins when it sinks the submarine or no such report goes out within 2 hours.
- **Convoy attack (7):** the frigate escorts a convoy of four merchants sailing east at 8 kn in a box about 1 NM around it. The submarine starts on the convoy's bow, about 10 NM ahead of it and 5 to 8 NM off its track. The submarine must sink two of them; only the mission submarine's torpedoes take a merchant (one hit sinks it) and each hit is logged. The orders give how many are sunk. The frigate wins when it sinks the submarine or the convoy holds out for 3 hours. Sinking a merchant here is the mission, not an incident.
- The submarine's end panel reads BROKE THROUGH, FRIGATE REPORTED or CONVOY HIT on a win, MISSION OVER when the time runs out.

## AI hunters

When nobody sails the frigate (the uConsole plays the submarine, or a solo browser plays the submarine), AI hunters crew every frigate station no browser holds; a station a browser takes is left to it at once. They read only what the frigate's own sensors report, never the submarine's position or identity:

- **Classification:** a contact whose heard signature the library knows only from submarines is classified submarine, as an operator comparing it with the library would; recognising it takes the operator 3 minutes on average. Other contacts stay unclassified.
- **Datum:** the freshest located submarine contact (ping, TMA or buoy fix); else the youngest of a radar mast track up to 10 minutes old, an HF/DF or ESM cross-fix up to 15 minutes old and an HQ submarine datum report up to 30 minutes old; else the bearing of a submarine contact; else the fresher of an HF/DF bearing and an ESM bearing on a mast radar up to 5 minutes old (an intercept the library matches to a submarine radar among its three best fits, with no radar or AIS ship within 10° of its bearing). The OPZ marks every bare radar blip of a raised mast or snorkel as a track, like an operator, and a mast track within 10° of a submarine contact's bearing counts as that contact. The radio room takes HF/DF bearings and cross-fixes like the autcrew. ELOKA plots an ESM bearing on a mast radar at most 30 s old as a line from the ship's position, a new line only after the ship has run 1 NM from the last one (at most 8, each for 15 minutes), and crosses the newest line with the latest earlier one it meets at 15° or more, no farther than 60 NM out: that ESM cross-fix is a position datum like an HF/DF fix.
- **Bridge:** in the convoy attack the frigate keeps station 3 NM ahead of the convoy, weaving 45° either side every 5 minutes (it closes at 18 kn when more than 2.5 NM off station), and prosecutes a datum only within 8 NM of the convoy. Otherwise, without a datum the frigate searches at 10 kn on a zigzag (legs of 10 minutes) whose base course turns 90° every 30 minutes. It runs at 18 kn to a position datum farther than 6 NM and works a closer one at 8 kn on a crossing course (60° off, switching sides every 5 minutes) so the towed array and TMA get bearing motion; on a bearing alone it steers 30° off it at 12 kn. It turns away from torpedoes and missiles like the autcrew and never steers into shoal water.
- **Sonar and weapons:** the ship pings once a minute on a submarine contact that has no fresh range, and fires one torpedo (or the set salvo) at a located submarine within 6 NM, again only when it has

stopped running. Nixies go out against a heard torpedo.

- **Helicopter:** launched for a datum within 30 NM (weather and deck permitting), after the deck has readied it (10 minutes on average); it flies to the datum, or 8 NM down a bearing, and dips. On a position it pings every 30 s; on a bare bearing it only listens. It drops a torpedo on a located submarine within 1.5 NM from a fix at most 2 minutes old, one at a time. Without a datum it recovers.
- **Patrol aircraft:** requested once a position datum exists (never for a bare bearing); it flies to the datum with its radar on, lays a circle of buoys where none listen within 4 NM, and attacks a located submarine within its drop range over the datalink from a fix at most 2 minutes old.
- **ASROC:** a position datum at most 2 minutes old from the ship's own sensors (not an HQ report or an ESM cross-fix) is passed over the datalink to the nearest friendly AI warship that carries ASROC and has it in range, at most every 2 minutes and never while an ASROC is in flight or its torpedo is running. The AI hunters do not fire the frigate's own ASROC or depth charges (those stay with a player at the Weapons station), and the submarine scenarios add no escort for it.
- The other stations (damage control, engine room, OPZ air defence, ELOKA) run the autocrew's policies. The hunt keeps no state of its own; the radar blips and the OPZ's marks it acts on (save v25) and ELOKA's ESM lines (save v39) are saved, so a loaded game continues it unchanged.

## Crew assist

Shift+F2 (on in a mission started from the multiplayer lobby with a browser taking part or with the uConsole host only) lets the AI man every station nobody holds, on the frigate and on a crewed submarine, so each player can stay on one station. A station a browser holds, and the one the uConsole shows, stay with their player; a station released in the browser ("Hand over to AI") goes back to the AI at once. The assist is saved with the mission (save v40).

- **Frigate:** the AI hunters above work the Bridge, Sonar, Weapons and the helicopter, the autocrew the other stations, also against an AI submarine.
- **Submarine command:** evades a torpedo or a ping it has heard, otherwise follows the submarine mission's leg or, in a frigate mission, closes a frigate the boat's own sonar has fixed within 12 NM and else patrols at 4 kn below the layer around its start point. It comes to snorkel depth when the battery falls below 35 % and nothing hunts the boat.
- **Submarine weapons:** keeps the tubes loaded, floods quietly once a heard target has a fix within 8 NM and fires one torpedo at a time down a fix within 4 NM. A target is a contact whose signature the library knows only from warships (in the convoy attack, from merchants).
- **Engine room:** snorkels to charge up to 95 % while unhunted, keeps the trim automatic, answers foul air with absorbers and oxygen candles and sends the two damage-control teams where fire, leaks or water are worst.
- **Sonar and mast:** the sonar keeps the focus on the loudest fresh contact; the mast comes down on such an alarm. Navigation and the radio room only keep watch.
- **A player's order wins:** a station the AI mans never overrides what a player at another station commands. With a player at Navigation the AI command leaves course, depth and evasion alone; with one in the engine room it leaves speed and silent running; with one at command the AI engine room leaves the trim and the damage-control teams, and the AI mast station leaves a mast raised at command or in the radio room up even on an alarm. While a player at the mast or in the radio room holds the mast up, the AI command keeps the boat at periscope depth; it dives again once the mast is down. On the frigate the AI Bridge does not steer while a player is in the engine room, and the AI weapons and patrol aircraft do not re-designate a current target a player chose at Sonar, OPZ or Weapons. A contact picked on the uConsole with Up/Down stays picked.

## Not modelled

- The radio room knows HQ's broadcast schedule, three kinds of order and situation reports: no free-text messages from HQ, no reception below 25 m (no ELF, no trailing wire), no burst transmission and no other units on the net; HQ's contact report is modelled intelligence, not a sensor of its own.

- Damage control is six compartments and two teams: no separate pressure-hull and outer-hull damage, no smoke or heat spreading, no fire in the air stores and no fire consuming oxygen; the submarine's overall damage (noise, top speed, sinking at 100 %) still adds up from hits, and the AI's submarines keep only that value.
- Below test depth the hull has no individual fittings, no gradual shrinking of the hull and no stronger welds from a refit; a failure picks its kind and compartment at random, and a crushed submarine is lost at once.
- The submarine's sounds are simple cues: stereo only tells port from starboard (ahead and astern sound the same), a detonation gives no range estimate and no creak comes from a particular compartment; the submarine's sonar room has no red light.
- The trim model is one weight and one moment: no free-surface effect, no compressibility of the hull with depth and no separate negative tank; the submarine does not surface fully. Food and fresh water do not run out.
- A raised mast or snorkel head is seen by the frigate's radar only as a bare blip (see the OPZ chapter). The AI submarine's periscope during its look round is found only by radar, never by the lookout's eye, and leaves no feather.
- The submarine's ESM hears no other submarine's radar and no missile seeker; it has no scored likelihood analysis, no scan-period measurement and no target motion analysis of an emitter (the cross-fix assumes a slow emitter).
- The fit grade is the crew's reading of the published ranges, not a likelihood: a wide-band radar measured near the middle of its range can fit better than the true emitter measured near its edge, and the browser shows the first 8 entries.
- The submarine campaign has no port choices beyond refit and quick turnaround, and no damage-control state carried over (only the overall hull damage). Outside the convoy attack the submarine's torpedoes home on the frigate only, and an AI submarine attacks merchants only as the convoy attack's mission submarine or as a patrol submarine of a frigate scenario.
- The AI hunters' ASROC comes only from friendly warships already in the scenario, never from the frigate's own launcher.
- The periscope has one magnification and no camera; sightings carry no identification beyond the coarse class, and the stadimeter assumes a class length rather than a masthead height.

## Mission and scoring

- Scenarios: 1 Patrol, 2 Double hunt, 3 Nuclear intercept, 4 Random (custom difficulty). User missions start from the Mission Editor (F5 in its browser). The runtime takes the editor's scope: a 500 NM fixed world or a packaged reference sector (sector:0 to sector:127), the authored weather, placed submarines, surface ships, aircraft (patrolling a 10 NM box at profile speed), animals, static decoys and hostile torpedoes already running on their course, seeded random groups, timed events (message, spawn, weather, objective) and the objectives sink, survive, protect (keep the named units alive until the time limit) and reach (enter the objective point's radius). Profiles saved in the Unit Editor can be placed like built-in ones and take effect in that mission (speeds, depth, torpedo load, behaviour, acoustics); a user submarine takes sensors, tubes, decoys and its battery, diesel or AIP plant from the built-in boat of its propulsion. Frigate and helicopter torpedoes, missing user profiles and other world sizes are refused at start. In the editor's World tab, Enter on Kind or Reference opens a pick list (Up/Down, PgUp/PgDn, Enter takes, Esc cancels); Reference lists the 128 sectors with their countries, and picking one makes the world a 500 NM reference world. The Preview tab then draws that sector's coast.
- Unit analyser (F8, main menu) and Unit Editor: the first page of every catalog profile in the analyser is a schematic 3D model of its type (tab 3D, then the sound and radar images with Left/Right); it turns slowly, and in the Remote Crew browser it can also be turned by dragging. The Unit Editor shows the same model under the selected profile and beside the fields of an opened one. Every ship, submarine and aircraft type of the catalog has its own model, built from the public main dimensions and general arrangement of the real class (Wikipedia; generic types such as a VLCC or a harbour tug use typical values): length, beam and draught, where bridge, masts, funnels, guns, missile cells, flight deck,

cranes and cargo stand, the submarine's sail, planes, rudders and missile deck, the aircraft's wings, tail and engines. The same type always looks the same; unit-editor profiles and anything without a type of its own keep the model of their class (warship, merchant, small craft, submarine, the lookout's helicopter). Torpedoes, decoys, whales, fish schools and jellyfish have their own models. The models are schematic and every one is drawn at the same length, so they are not to scale with each other. The same models stand in the eyepieces, turned by the judged angle on the bow (lookout, binoculars, periscope); the browser loads them in three groups on first use.

- Win: all targets sunk, or survive the time limit. Lose: own ship sunk, civilian hit, target 150 NM from its start, or time out.
- Score: 1000 per sunk submarine, 200 per unused torpedo, 500 without civilian losses, up to 500 time bonus.
- Debrief: after the mission ends, D on the end panel opens the debrief. It replays the mission with the truth beside what the crew knew: the true tracks of the ship and the hostile submarines, the crew's contacts where it had placed them (bearing-only contacts as bearing lines), weapons, buoys and aircraft. Beside the chart it lists the time of the first contact, first fix and classification, weapons fired and submarines sunk, the mean error of the crew's fixes, and every event; a "missed chance" is a hostile submarine within 4 NM for at least 5 min without any contact, marked above or below the layer. Left/Right step (Shift: 1 min), Up/Down or PgUp/PgDn jump between events, a click on the timeline jumps there, D or Esc returns. The debrief is recorded every 10 s (coarser on long missions), is never shown during a mission and is not saved: after a load it covers the mission from the load onwards.

## Glossary

Term	Meaning
ASW	Anti-submarine warfare
HMS / TAS	Hull-mounted sonar / towed array sonar
VDS	Variable-depth sonar: towed body lowered below the layer
LOFAR	Low-frequency analysis and recording: frequency-time waterfall
DEMON	Demodulated noise: reveals blade and shaft rate
TMA	Target motion analysis from bearings
BT	Bathythermograph: measures the sound-speed profile and layer
CZ	Convergence zone
Datum	Last estimated target position used for weapons and search
CIC / OPZ	Combat information centre / Operationszentrale
ESM / ECM	Electronic support (passive intercept) / countermeasures (jamming)
HFDF	High-frequency direction finding
EMCON	Emission control: radars off
ROE	Rules of engagement

## Screen abbreviations

When a full label does not fit the 1280x720 screen, the station shows its catalogue abbreviation instead of cutting the text off. The F11 log and tooltips always show the full wording.

Short	Meaning
CRS/SPD	Course / speed
NOISE, CAV	Own radiated noise in %, cavitating
SS/LAYER	Sea state / measured layer depth (BT)
FLOOD	Mean flooding
TORP, VLS/CHAFF	Torpedoes left, VLS cells / chaff reload
HELO/ROE, HGR	Helicopter state / rules of engagement, hangar
BRG, G, N	Bearing, gain, notch
BB, FILT, HET	Broadband, filtered, heterodyne audition
STOW, DEPLOY, RECOV, OUT, STAB	Towed array stowed, deploying, recovering, streamed, stability
RDY, N/RDY	Ready, not ready
UNK, FRD, NEU, HOS	Affiliation: unknown, friend, neutral, hostile
SFC, SUB, AIR, MSL, TRP	Domain: surface, subsurface, air, missile, torpedo
RDR S/A	Radar surface / air
D, W, CD	Dipping sonar depth, water depth, ping cooldown
B-rate, Tgt	Bearing rate, target